TECHNISCHE UNIVERSITÄT DRESDEN

KAAI: Knowledge-aware Artificial Intelligence

Faculty of Computer Science
Institute for Artificial Intelligence
Chair of Knowledge-aware Artificial Intelligence

Master Thesis

Illustrating GenAI knowledge bases with GenAI vision models

Stella Brinster
Born on: 04.12.2001 in Freiberg
Matriculation number: 4876386

October 28, 2025

First referee
Prof. Simon Razniewski

Second referee
Prof. Michael Färber

Supervisor
Prof. Simon Razniewski

Statement of authorship

I hereby certify that I have authored this document entitled *Illustrating GenAI knowledge bases with GenAI vision models* independently and without undue assistance from third parties. No other than the resources and references indicated in this document have been used. I have marked both literal and accordingly adopted quotations as such. There were no additional persons involved in the intellectual preparation of the present document. I am aware that violations of this declaration may lead to subsequent withdrawal of the academic degree.

Dresden, October 28, 2025

Stella Brinster

TECHNISCHE UNIVERSITÄT DRESDEN

ScaDS.AI DRESDEN LEIPZIG

Faculty of Computer Science Chair of Knowledge-aware Artificial Intelligence

Master Thesis

Illustrating GenAI knowledge bases with GenAI vision models

Stella Brinster

Born on: 04.12.2001 in Freiberg
Matriculation number: 4876386

October 28, 2025

First referee
Prof. Simon Razniewski

Second referee
Prof. Michael Färber

Supervisor
Prof. Simon Razniewski

Abstract

Knowledge bases (KBs) and knowledge graphs (KGs) have long been created and maintained through human labor. While this ensures high-quality data, it makes maintenance and expansion very costly and time-consuming, and invites content gaps. One of the first gaps to be created is usually the missing multimedia data, since it is difficult to gather or create usable data by hand. Because of this, this work utilizes GenAI in an approach based on Abu Ahmad et al. (2023) to generate representative images for a KB. To, at the same time, further add to research on the internalized factual data of large language models (LLMs), the KB chosen to illustrate was GPTKB, a KB generated by an LLM without external knowledge. This thesis uses Llama-4-Scout-17B-16E-Instruct to construct natural language prompts from all triples within a subject's GPTKB entry. These text prompts are fed into the Stable Diffusion model, stable-diffusion-3.5-large-turbo, to generate images with the potential to represent the subjects in their GPTKB entries. Metrics are applied to assess the feasibility and adequacy of this work's approach, including the quality of the generated images' visual attributes, their appropriateness within the semantic context of the subject they illustrate, the presence of noticeable biases, and the occurrence of logical errors. This work also provides a generated set of over 100,000 images illustrating subjects of GPTKB 1.5.1 and the prompts used to generate them.

Contents

1 Introduction

This chapter provides an introduction to the thesis, outlining the motivation for this work, the specific problems it addresses, and the contributions it makes.

1.1 Motivation

With the increasing use of artificial intelligence (AI), knowledge bases have become more important as well (McGrath (2024), Hajkowicz et al. (2023), Ding et al. (2025), Fan et al. (2024)). However, not just AI models use these extensive databases for learning and information lookup, so do humans. Contrary to computational models, many humans derive significantly more value from visual than from written data, especially regarding memorization and recognition. This phenomenon is often referred to as the "Picture Superiority Effect" (Hockley (2008), Beagle (2009), Childers and Houston (1984), Stenberg (2006)).
Since visual data is more challenging to collect and moderate, human-edited knowledge bases often lack it. As generative artificial intelligence (GenAI) has risen in popularity and image quality, using it to create more visual data easily seems like a logical step. This work aims to develop an approach to generate images from knowledge base entries, at a scale large enough to add meaningful value to knowledge bases. Fig. 1.1 shows a comparison of a real image (Fig. 1.1a) and a generated image (Fig. 1.1b), and the prompt that generated that image (Fig. 1.1c) for the subject "Apostolic Palace".

In this process, it will address the challenges of this endeavor: selecting a vision model to use, selecting a prompt phrasing and prompt knowledge source to generate images, and assessing the feasibility of including long-tail subjects. Furthermore, this thesis involved itself with well-known issues of vision language models (VLMs), like Dall-E 2 (Ramesh et al. (2022)), Midjourney (Midjourney (2022)), or Stable Diffusion (Rombach et al. (2022)), which have high costs in comparison to large language models (LLMs) and significant biases. In addition to these process- and model-dependent challenges, the sheer size of knowledge bases adds to the problems this work seeks to address, aiming to find a scalable approach to automatically illustrate knowledge bases.

To illustrate a knowledge base, an easily accessible knowledge base was needed. Due to its free accessibility and the absence of multimedia entries, GPTKB (Hu et al. (2025a)) was chosen. Since it was purely generated by GenAI and aims to provide information about the knowledge that can be extracted from GenAI alone without external sources, GPTKB further aligns with the intention of illustration through GenAI. Illustrating GPTKB through GenAI serves this purpose and could further contribute to future studies in AI-inherent knowledge by analyzing the resulting images.

1.2 Problem Statement

Three main points have been considered in this work: the generation method, the approach's scalability, and the implications and challenges. Each can be split into two major questions, creating the following six research questions. From this point on, they will be referred to with their corresponding shorthand RQ 1 to RQ 6.

RQ 1: Does the chosen model substantially impact the results?

RQ 2: Which method of generating prompts is best suited for illustrating knowledge base entries?

RQ 3: Does the image adequacy suffer when entities become less well-known?

RQ 4: Can images be reused for related entities without drastically reducing the image adequacy?

RQ 5: Do the generated images add significant worth to knowledge base entries in comparison to triples?

RQ 6: Are there notable biases in the generated images?

1.3 Contributions

The main contribution of this thesis is the in-depth study of the six research questions RQ 1 to RQ 6. In particular, it was found for RQ 1 that, while the model affects image adequacy and quality, the main differences between models lie in pricing and the set of filters for prompts. For RQ 2, it was determined that KB prompts created from triples extracted from

(a) Image of "Apostolic Palace"; as found at Vatican Tickets (2025)

(b) Image of "Apostolic Palace" generated through Stable Diffusion

```
A highly detailed and realistic depiction of the Apostolic Palace, gptkbp:instance_of:palace, located in Vatican City, showcasing its stunning Renaissance architecture and ornate decorations, set against a serene backdrop of Vatican Hill.
```

(c) The prompt that generated the shown image in 1.1b

Figure 1.1: Comparison of real and generated image, plus the prompt for the generated image

a subject's knowledge base entry and reformulated into natural language prompts were the best choice. Furthermore, while exploring RQ 3, it was found that less well-known, so-called long-tail, entities did not seem to generate significantly less adequate images. Regarding RQ 4, this thesis has found that the generated images add about as much value as half of a randomly chosen predicate-object-tuple in a knowledge base entry. Lastly, regarding RQ 6, it was determined that the generated images do hold notable biases compared to the original knowledge base, especially regarding gender.

An overview of the approach investigated by this thesis can be seen in Fig. 1.2. In general, this work aims to propose a continuous cycle of receiving unillustrated subjects from GPTKB, creating prompts for them based on their entries, and generating representative images to add back into GPTKB. This would allow continuous updates to the image-less entries of GPTKB until all of them are illustrated. Reuse of already generated images was tested and is recommended for future, large-scale use, though it was not fully implemented in this work. It was determined that entries of subjects with generated illustrations should be compared with those of subjects yet to be illustrated. This comparison was found to be easiest by creating text embeddings for the entries once and comparing these embeddings through a model like Sentence Transformers (Reimers and Gurevych (2019)).
The additional tests done during this thesis, using CLIP and UQI scores to find the best-suited model and prompting approach by judging image quality (visual characteristics of the image, like blurriness or contrast) and adequacy (how well an image suits a subject and its entry), can be omitted for future use, unless a new determination concerning these factors becomes necessary.

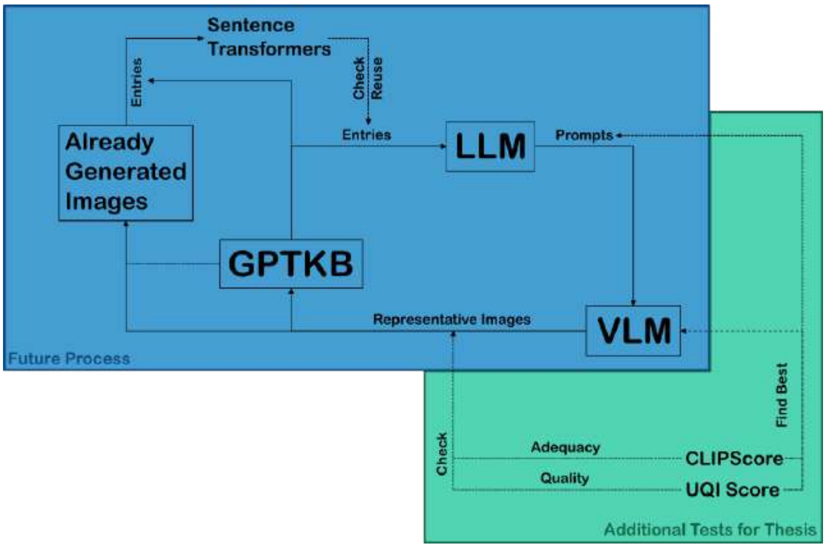


Figure 1.2: Schematic of the processes done and intended for the future by this thesis

This work produced a Python script that included both prompt generation from knowledge base entries and image generation from those prompts. The prompt generation was developed through various tests regarding model choice, entity popularity, reuse, and image quality, trying to optimize the results. With that, 104,611 images for the entries of GPTKB were generated, and the approximate worth of an image in comparison to triples was determined. Lastly, the generated images were checked for biases, and the results were documented. All important scripts, as well as most generated prompts and images, can be found in its GitHub repository at https://github.com/Knowledge-aware-AI/GenAI-KB-illustration.

2 Related Work

In this chapter, the basic concepts used in this thesis will be outlined through related works. Specifically, generative AI and knowledge bases will be looked at in detail.

2.1 Generative AI

In recent years, generative AI has become so popular that large language models (LLMs) like GPT are used as AI assistants in daily life. The following provides a short introduction to GenAI as a whole and to vision models specifically, which played a major role in this work.

2.1.1 Overview

AI in general has been around for quite some time (Hajkowicz et al. (2023)), but generative AI, or GenAI for short, specifically, has taken off in recent years (McGrath (2024)). While traditional AI was able to seemingly "learn", too, it is still following an algorithm. Comparing the two is akin to comparing someone following step-by-step instructions for building a table with someone using general woodworking principles to create a new table that has not been made before: Both end up with a table, but one of them is more uniquely theirs than the other. The solutions GenAI comes up with are similar to how and what a human might come up with, learning from preexisting material to create entirely new things from that (Abukmeil et al. (2021), Gui et al. (2020)).
It simplifies the creation process for whatever it is trained to generate – text, illustrations, diagrams, code, or even music – be it for research or personal use. This means that the required artistic skill to create high-quality art is reduced, making it more accessible to anyone (Ye et al. (2024)). Most models can be used through text prompts. A simple, short sentence can create an entire image by using a Stable Diffusion model (Rombach et al. (2022)) or DaLL-E 2 (Ramesh et al. (2022)). Text creation is often done by large language models like GPT (OpenAI et al. (2024), OpenAI (2025)) and Llama (Touvron et al. (2023)), which can generate human-like natural language. They show impressive reasoning ability and can have full, logical conversations.

Only recently has research and usage of GenAI skyrocketed, with a transformative impact on AI research as a whole. Not only has it become a part of many people's personal lives with AI assistants like ChatGPT (OpenAI (2022)), Gemini (Team et al. (2025)), or Claude (Anthropic

(2023)), but its usage in large companies and research has risen starkly as well (Andersen et al. (2025), McGrath (2024)). Research on GenAI has increased exponentially in recent years, underscoring its impact on the research landscape (Ding et al. (2025), Obreja et al. (2025)).

2.1.2 Vision Models

Vision models are a part of GenAI. As the name suggests, they specialize in generating visual data such as images, illustrations, graphs, and videos. Graph creation is mainly sought for in research to more easily create meaningful visualizations of data, while the generation of more natural images and illustrations is primarily of interest for personal use. In both use cases, image quality is essential. For graphs, the accuracy and integrity of the portrayed data should be the major focus (Ye et al. (2024)), whereas natural images place greater emphasis on aesthetics and composition, as long as the central theme or focus is correct.

Vision language models (VLMs) use user-generated text prompts to generate visual representations. The most prominent models currently on the market are GPT-image-1 (Open AI (2025)), Dall-E 2 and 3 (Ramesh et al. (2022), Betker et al. (2023)), Stable Diffusion models (Rombach et al. (2022)), like Stable Diffusion 3.5 Large Turbo (Stability AI (2024a)), and Llama 4 (Meta (2025a)). All of them can be considered state-of-the-art VLMs, though information about them is not equally available. While the parameter counts for Llama (Llama-4 Scout 17B 16E Instruct: 109 billion parameters, 17 billion activated; Meta (2025b)) and Stable Diffusion (Stable Diffusion 3.5 Large Turbo: distilled from 8.1 billion parameter Stable Diffusion 3.5 Large; Stability AI (2024b)) are entirely or at least in part disclosed, the counts for GPT-image-1, Dall-E 2, and Dall-E 3 can only be estimated. However, even just these two numbers clearly show how different model sizes can be. Commercial models like GPT-image-1, Dall-E 2, and Dall-E 3 offer API-based image generation at different price points. A brief overview of their pricing is presented in Tab. 2.1 to give an idea of their costs as well as compare them to costs for an LLM (GPT-5, OpenAI (2025)).

Model	GPT-image-1	Dall-E 3	Dall-E 2	GPT-5
Worst Quality	Low	Standard	Standard	-
Smallest Size	1024x1024	1024x1024	1024x1024	-
Price per Image	$0.011	$0.04	$0.016	-
Best Quality	High	HD	Standard	-
Biggest Size	1536x1024	1536x1024	1536x1024	-
Price per Image	$0.25	$0.12	$0.02	-
Input[1]	$5	-	-	$1.25
Cached Input[1]	$5	-	-	$1.25

Table 2.1: Price ranges for commercial models by OpenAI

In addition, GenAI in general has two large issues: biases and hallucinations. Biases influencing results, both ex- and intrusively, can be seen in both LLMs and VLMs and have been studied and surveyed many times before (Wei et al. (2025), Park (2024), Gallegos et al. (2024), Sakib and Bijoy Das (2024)). While the biases often veer into topics of amplifying stereotypes (Bianchi et al. (2023)) as well as overrepresenting certain genders and ethnicities (Zhou et al. (2024)), the hallucinations usually refer to the models inventing data that was never within training data and is not factually accurate (Aithal et al. (2024), Fan et al. (2024)). In image

[1] Price per 1M tokens

generation, such hallucinations usually manifest as extra limbs, object or animal fusion, or discontinuities in items like railways (Aithal et al. (2024), Ma et al. (2025), Fu et al. (2025)).

2.2 Knowledge Bases

As this work mostly aims to illustrate knowledge bases (KBs), this section will explain knowledge bases and their uses, multimedia data in connection with knowledge bases, and the knowledge base used for this work, GPTKB.

2.2.1 Foundation

Training AI, especially large language models, requires vast amounts of preexisting data. Because of that, general domain knowledge bases such as Wikidata (Vrandečić and Krötzsch (2014)), Yago (Suchanek et al. (2007)), or DBpedia (Auer et al. (2007)) have long become indispensable resources for the creation of AI applications. These extensive, structured collections of information are still primarily created, edited, and moderated manually, or through scraping semi-structured web content (Weikum et al. (2021)).
This makes their upkeep and expansion both expensive and labor-intensive. While this ensures high-quality information and a clean structure, the rate of expansion is naturally slow. This invites content gaps, leading to incomplete entries of varying shapes. One of the parts especially affected by these content gaps is multimedia data.

Tab. 2.2 shows an excerpt from the knowledge base entry of the item "Apostolic Palace" (Q145093) on Wikidata (Wikidata (2025)).

Property	Item	References
instance of	palazzo	0 references
	palace	1 reference
	palace of the Popes	0 references
	official residence	0 references
inception	30 April 1589 (Gregorian)	0 references
has use	official residence (used by pope)	0 references
	museum	0 references

Table 2.2: First 7 statements for "Apostolic Palace" (Q145093) from Wikidata (2025)

A knowledge graph (KG) is most often defined as $\mathcal{G} = \{\mathcal{E}, \mathcal{R}, \mathcal{T}\}$. This can be understood as a knowledge graph $\mathcal{G}$ consisting of a set of entities $\mathcal{E}$, a set of relations $\mathcal{R}$, and a set of triples $\mathcal{T}$. The triple set $\mathcal{T}$ consists of a head entity h, a relation r; $r \in \mathcal{R}$, and a tail entity t. For this, $h, t \in \mathcal{E}$ applies. Compared with the KB entry example in Tab. 2.2, the structure can be easily recognized. Wikidata, as a knowledge base, is also widely recognized as a knowledge graph, due to its RDF-triple structure, which naturally builds a graph. It should be noted that, while all knowledge graphs are knowledge bases, not all knowledge bases are necessarily knowledge graphs. However, KBs commonly used in AI research, such as Wikidata, Yago, or DBpedia, are often KGs, as their graph structure is advantageous for such projects (Paulheim (2016)). If a knowledge graph has additional multimedia data, it is considered a multi-modal knowledge graph (MMKG). With the rise of GenAI, the idea of generating missing image data to transform existing KGs into MMKGs has come up. The issue with this approach is that it usually requires manually-orchestrated prompts to ensure high image quality.

Abu Ahmad et al. (2023) previously used triples from DBpedia to generate images for subjects, proposing a possible way to automate this process. Xu et al. (2025) has built upon this, creating a neighbor search to add more valuable entities into the prompts. So far, neither of them has attempted to actually scale their processes closer to the order of magnitude of full knowledge bases. Abu Ahmad et al. (2023) did their research on a set of 1500 fictional characters with images and 1500 fictional characters without images, and Xu et al. (2025) used two datasets, one based on Wikidata with 14244 images and one based on DBpedia with 12818 images. This thesis aims to increase those numbers, actively scaling the process to generate at least 100,000 representative images.

2.2.2 Multimedia Data

Multimedia data refers to data that combines at least two different types of data. Since text data is the basis of knowledge bases, multimedia data in this work refers to all other types of data, like images, videos, or audio. All of these are rather scarce within knowledge bases. Generally, images are the most common type of data found in knowledge bases, in addition to text, and are also the simplest to generate currently. Because of that, this work focuses solely on increasing the amount of images in a knowledge base, though the approach itself could be adapted for other data types as well.
Wikidata, as a well-known and very large knowledge base, counts 118,981,892 items in its statistics.[2] A query on Wikidata counts 5,981,073 images overall, and 5,797,751 items with images.[3] This means that only around 5 %[4] of items on Wikidata have images in their entry. This is an example of how severely lacking knowledge bases truly are when it comes to images. Since visual data – like images, illustrations, and graphs – can encode information that is difficult to store in natural language, reducing this content gap would be beneficial – Adding images to a knowledge base manually, however, holds many challenges. Copyright and other licenses need finances, and making sure the image is suitable takes time. With the rise of GenAI, however, the question arose whether images could simply be generated to fill these gaps, if only as placeholders until manually compiled images can be added. To test the utility of this approach, images were generated for GPTKB as a freely usable knowledge base.

2.2.3 GPTKB

To reduce the enormous financial and labor costs associated with the creation of knowledge bases, as well as to gain insight into the knowledge and beliefs of language models, GPTKB was created. It is a pilot project in which a knowledge base was created through recursive graph exploration. Its website states that this process reduced the costs by 10 times in comparison to other knowledge base creation projects, while creating 100 million triples for more than 6.1 million entities as of the latest release, version 1.5.1[5] (Hu et al. (2025a)). As an example, the entry for the same entity[6] as shown in Tab. 2.2 is shown in Tab. 2.3.

Through its demonstration of large-scale knowledge base construction by an LLM, GPTKB is a milestone for natural language processing and the Semantic Web alike. It allows a look

[2] As stated in the statistics on 19.09.2025.
[3] The queries were run on 19.09.2025, with 'hint:Query hint:optimizer "None".' to avoid timeouts.
[4] 4.87 %
[5] As of 19.09.2025
[6] As of 23.10.2025, the original subject "Apostolic Palace" has been removed or renamed from the available GPTKB-website (GPTKB version 1.5), so the most similar entity "Vatican Apostolic Palace" was chosen.

Predicate	Object
gptkbp:instanceOf	gptkb:palace
gptkbp:builtIn	16th century
gptkbp:function	official residence of the Pope
gptkbp:usedFor	gptkb:Pope administrative offices official ceremonies Papal audiences

Table 2.3: Tuples for "Vatican Apostolic Palace" from GPTKB (2025) mirroring statements shown in Tab. 2.2 as closely as possible

into the inner beliefs of LLMs about how entities are connected and opens a new path in the field of general-domain knowledge base creation (Hu et al. (2025b)). Furthermore, the current research into GPTKB was able to pinpoint certain challenges of this approach, such as entity recognition, entity and property canonicalization, and taxonomy construction (Hu et al. (2025a)).
This work used GPTKB versions 1.1.1, 1.5, and 1.5.1, the latest of which was used for the large-scale generation, yielding over 100,000 images.

3 Methodology

This chapter will describe the datasets and approaches used in this work. It will report the datasets and evaluation metrics, as well as the approaches regarding the investigations of the generation method, scalability, and utility and bias utilized in this thesis.

3.1 Datasets

Within this work, five datasets have been used: Regular, Random, New Random, Scale Test, and Large Scale. All of them have been derived from GPTKB in some way, from either version 1.1.1 or version 1.5.1. The datasets Regular, Scale Test, and Large Scale also had subsets or modified versions made: Reuse Company/City and Reuse Product/Landmark for Regular, a randomly chosen 500-subject subset for Scale Test, and a 500-subject, persons-only subset for Large Scale.
The dataset used for the initial tests, referred to as "Regular" from here on out, consisted of 120 subjects. They were handpicked from GPTKB with the following criteria:

- There have to be 20 subjects each for the six chosen categories: people, cities, landmarks (Tab. 3.1), companies, products, and book series (Tab. 3.2).
- Each product has to correspond to a company, and each landmark has to correspond to a city.
- The people have to be split into 10 well-known and 10 lesser-known persons.
- The cities should be split into 10 western and 10 non-western cities, which are each split into 5 small (western: < 500,000 inhabitants, non-western: < 3,500,000 inhabitants) and 5 large cities each.
- Every subject needs to have an entry in GPTKB (version 1.1.1) and an image on Wikidata to use as a ground-truth.

For the next tests, a randomly chosen dataset was used to include even more long-tail entities, from this point on referred to as "Random". 120 subjects were chosen randomly and without any filter from GPTKB (version 1.1.1). To better compare them with the Regular dataset, the chosen subjects were split into categories with the help of ChatGPT (GPT-4). For that, ChatGPT was given the task to minimize the number of categories and keep them as equally sized as possible. This sorting was checked manually and refined as needed, adding not included subjects into the categories and removing duplicates.

People	Cities	Landmarks
Maya Widmaier-Picasso	Vatican City	Apostolic Palace
Edgar Allan Poe	Miami	Freedom Tower
Garry Kasparov	Vienna	St. Stephen’s Cathedral
Mary Wortley Montagu	Columbus	Scioto Mile
Johann Wolfgang von Kempelen	San Marino	Guaita Fortress
Wright brothers	Kansas City	The Folly Theater
Albert Einstein	Sacramento	Tower Bridge
Robert Knox	Fresno	Fresno Water Tower
Hitler	Amsterdam	Dam Square
Ada Lovelace	Brussels	Manneken Pis
M. Carey Thomas	Manama	Bahrain Fort
Roberta Wright McCain	Rabat	Mausoleum of Mohammed V
Piotr Hofmański	Kuala Lumpur	Chinatown
Linus Yale Jr.	Manila	Intramuros
Bessie Blount	Dubai	Burj Khalifa
Pedro Paixão	Nairobi	Bomas of Kenya
William C. Gorgas Jr.	Singapore	Botanic Gardens
Richard Fleeshman	Bangkok	Central World
Rushanara Ali	Dhaka	Jatiyo Sangsad Bhaban
Zorion Eguileor	Cairo	Al-Azhar Mosque

Table 3.1: Subjects of Regular dataset for the categories: People, Cities, and Landmarks

Companies	Products	Book Series
The J. M. Smucker Company	butter	Miss Peregrine’s Peculiar Children
Fujifilm Baltic SIA	film	Dummies series
Suncor Energy	natural gas	Discworld
Symington Family Estates	port	James Bond Novels
United States Rubber Company	footwear	Wen Xuan
Snap Inc.	Bitmoji	The Secrets of the Immortal Ni-
Johnson & Johnson Con-	health services	cholas Flamel
sumer Health	medical devices	Bridgewater Treatises
B. Braun Melsungen AG	canvas	Les Rougon-Macquart
Daler-Rowney	sunglasses	The Art of Computer Programming
Luxottica Group	Unity	Principia Mathematica
Unity Technologies	video games	Winnetou
Marvelous Entertainment	tobacco	Jip and Janneke
American Tobacco Company	Electron rocket	McGuffey Readers
Rocket Lab Ltd.	UXPin	Voyages Extraordinaires
UXPin, Inc.	music	American Guide Series
Sony Corporation	sensors	Lang’s Fairy Books
Tyco Electronics	accessories	The Rover Boys
Tory Burch LLC	clothing	Doctor Dolittle
Gap, Inc.	Telecommuni-	Harvard Classics series
Furukawa Co., Ltd.	cations	The Witcher

Table 3.2: Subjects of the Regular dataset for the categories: Companies, Products, and Book Series

Randomly Selected Subjects		
The Museum of Contemporary Art in Bakersfield	Filippo Strozzi the Younger	German Freikorps
Bernhard of Carinthia	Boris Hybner	Matsuyama City Museum of Poetry
Bahawalpur Technical Institute	Brigadier General Janis Sprogis	Ge Force GPU
Vikram Aur Betaal	Agnès Baltsa	Ajman Free Zone Authority
Capture of Fukuoka Castle	Ms. Pat	Rally of Czech Republic
Thomas Saltonstall	Second Anglo-Boer War	Algari Rose
Churfirsten	Henry Fairfax, 2nd Lord Fairfax of Cameron	Murakami vs. Murakami (2016)
Governor of Colorado Territory	Riga’s Historic Centre	K9999
The Binti Universe	Detective Philippe Rousselot	Cherokee Nation v. United States Department of the Navy
And Then We Danced	Un Verano Sin Ti (Deluxe)	Dr. Richard N. Longenecker
U. N. forces	Kordes’ ’Sunsprite’ rose	Chico the Jet Hawk
H. C. Mc Culloch	Prabhat Prakashan	Paligo
Gisele Yashar	Radomir Putnik	Seguin Transit
The presidential seal of Mexico	Dennis the Menace Strikes Again	Cagayake! GIRLS
Captain America #42	NBC Sports Washington’s The Game Plan	Zalaegerszeg Adventure Park
George Burnham Jr.	Valkenburg railway station	Great River Station
Palazzo Ca’ Corner della Regina	Molesworth Airfield	Eddie Goldman
Bloomsbury Publishing	General Nguyen Van Hinh	Ammonius Hermiae
Grenada	Busan Sajik Stadium	Cott Canada
All Blacks Sevens	Ceres, California	567 U. S. 709
The Immortal Hulk Vol. 10	Lizarazu	Mary Fitz Roy, 4th Duchess of Richmond
Pingtung County Government	Molly Meldrum Foundation	Symphonic Live
Kintai Bridge Park	Sheffield to Brora Line	Stila
The Merchant of Venice (Broadway)	Joseph Sewall III	Malden City Hall
Ascanian coat of arms	Lemaître’s redshift	Piano Sonata No. 19 in C major, Op. 2
Ban Chiang archaeological site	Tokyo Metropolitan University Athletic Association	Brit Asia TV Music Awards
Battle of Håkonarstadir	Paramount D. Smith	Liberty Global Partnerships
Erfurt Hauptbahnhof	Wizarding Wireless Network	Sri Bhashya
Fresco of the Sacred Snake	Sérgio Reis -Ao Vivo (2015)	The Benin ceremonial drum
Yokohama Hakkeijima Sea Paradise	Baron of the Isle of Benbecula	Tlaquepaque
FENSTERBAU FRONTALE	29th Logistics Support Battalion	Puzzle Bobble: Infinity
Congo Free State Administration	Rurouni Kenshin: Kengeki Kenran 26	LCD Soundsystem -This Is Happening
KSL City Mall	Flair for Fashion	Dodge Dart Rallye
Best Music at the BAFTA Games Awards 2016	The New England Transcendentalists	The Style Council Big Band
Screamfest Horror Film Festival 2008	Bell Productions	Richard Day
Battle of Yavin	Giorgio Graziani	Michele A. Lee
Guillaume, Count of Flanders	Hiroshima University School of Dentistry	Marta Lobo Antunes
Karpov vs Korchnoi, Game 19	1958 Baghdad Pact withdrawal	The Battle of the Shining Walls
	St. Albans City School District	Massachusetts Route 9 G
	Principality of Antioch	George M. Marakas
		Vladimir Stimac
		Perlman Plays Tchaikovsky
		UN Security Council Resolution 1704

Table 3.3: Subjects of the Random dataset

However, this manual refining step was not used to check the correctness of the assignment to the categories, as that would have been very time-consuming. All chosen subjects can be found in Tab. 3.3. The categories, as well as which subject was assigned to which category, can be found in Tab. 6.1 in the appendix. Another dataset was created by choosing 200 new, random subjects from GPTKB v1.5.1 in the same way as the Random dataset. This dataset will from now on be referred to as "New Random". It was sorted into categories by ChatGPT as well, and the sorting can be found in Tab. 6.2 in the appendix.

For each of the total 440 chosen subjects across the Regular, Random, and New Random datasets, three levels of knowledge source of prompts were generated, as well as an image for each prompt type. The knowledge sources will be referred to as VLMp, LLMp, and KBp, and will be addressed in further detail in Subsection 3.3.2. The scalability test and the final generation, Scale Test and Large Scale, used neither of these datasets. Instead, a prompt type was chosen, and the images were generated for the subjects in GPTKB v1.5.1 in the order they were saved there. The scalability testing dataset included 5000 subjects and will be referred to as "Scale Test". It was sometimes split into a randomly chosen subset of 500 unique subjects. Finally, a dataset consisting of 104,611 images was created in the same manner as the Scale Test, further referred to as "Large Scale". It was also split into a 500-image, person-only subset used for bias checks.

3.2 Evaluation Metrics

To evaluate the adequacy and quality of the generated images objectively and answer the research questions RQ 1 to RQ 6, metrics had to be chosen. Different types of preexisting metrics were considered, differentiating them by ease of use and the amount and usefulness of the information they can provide. In the end, it was decided to use two of the scores previously used by Abu Ahmad et al. (2023), which this work is heavily based upon. The FID score they used was tried, but dropped as it did not seem to provide much readable insight regarding the representativeness of the images. Both other scores they used, the universal image quality index, UQI (Zhou Wang and Bovik (2002)), for the general quality of an image, and the CLIPScore (Hessel et al. (2021)) for the perceived adequacy of an image for a subject, were used in this work. They, as well as a guessing game to further investigate the recognizability of subjects from their images, will be further detailed in this section.

3.2.1 UQI Score

To calculate the UQI scores in this work, the easy-to-use Python library "sewar" was used (Khalel (2025)). The image to judge, as well as a reference image, was given to a function to calculate the pair's UQI score. Since UQI needs the images' sizes to match up for the pixel-by-pixel comparison, the reference image was scaled and cut to the same size and aspect ratio as the generated image.
The UQI (universal image quality index) is a pixel-based metric comparing the similarity of the actual pixel-by-pixel image to a reference image of the same size. Because it just compares individual pixels with each other, it does not judge the semantic context of images or whether they show the same subject, but instead simply looks at how similar the pixels of an image are to those of a reference image. It is bounded between -1 and 1, where 1 means identical images, 0 means the images show no correlation at all, and -1 means the images show negative correlation (Zhou Wang and Bovik (2002)).
Since UQI only compares the actual pixel values of images, all semantic context regarding the

portrayed entity is missed. Instead, it compares general image attributes like the distribution of contrast, brightness, and color. Therefore, UQI cannot help to assess the adequacy of an image for its subject, but it can be used to filter out very bad images with major blur or random, single-color splotches. Furthermore, the negative correlation only relates to strictly linear negatives, so negative values in general are not expected with natural images. For example, the UQI score for the image of a cat and the same image is 1; if compared to the same image with inverted colors (photographic negative), the UQI is approximately 0.40; and if compared to a solid black image of the same size, the UQI is 0. If the cat image is compared to the image of another cat, the UQI score reaches around 0.43. Comparing the image of a cat with the image of something completely unrelated and different, like a tree canopy, the resulting UQI score lies around 0.53. As seen from this, the UQI score does not judge the subject of an image at all.

3.2.2 CLIPScore

The CLIPScores in this work were calculated through the Python library "torch" (Ansel et al. (2024)) in addition to "transformers" (Wolf et al. (2020)). For that, a function was handed both the generated and the ground-truth image to calculate the CLIPScore for the pair. Since CLIPScore compares embeddings and not pixel-by-pixel, different sizes of images did not matter, and the images were not edited further before score calculation.
The CLIPScore is an evaluation metric based on the CLIP (Contrastive Language-Image Pre-Training) neural network. This model has been trained on text-image pairs and has been found to reliably judge how well an image and prompt fit together, closely resembling human judgment (Hessel et al. (2021)). The CLIPScore metric is defined as:

$$\text{CLIPScore(I, C)} = max(100 * cos(E_I, E_C), 0)$$

It is equivalent to the cosine similarity between the embedding of an image E_i and the embedding for the text E_C for a caption C. Since CLIP calculates the similarity between embeddings, it can do so as well for image-image pairs, by exchanging the text embedding E_C with the embedding of a reference image.
The nature of the CLIPScore calculation keeps the resulting score bounded between 0 and 100, with 100 as a perfect match. Matching the descriptions in Hessel et al. (2021), the CLIPScores calculated in this work stayed closer to 30 for text-image pairs, while they were around 80 to 90 for image-image pairs. While the reference-less method of comparing the prompt with the image was attempted, comparing images yielded better results for this work's intentions. The scores for image-image comparisons are the only ones reported in this thesis. As a reference point, the score of a random cat image with itself is 100; scoring it against a completely black image yields a score of 88.3; scoring it against a completely unrelated image, like a tree canopy, yields 89.7; and comparing it to the same image of a cat with inverted colors (photographic negative) gets a CLIPScore of 97.2. If compared to the image of a different cat, the resulting CLIPScore is 95.4. This means that scores below 90 point at unrelated image subjects, while very closely related images will most likely have a score above 95. For the datasets Regular, Random, and New Random, the reference images were collected manually. For the Scale Test dataset, the reference images were collected by crawling Google Images through the Python library "icrawler" (Chen (2025), Chen (2018)).

3.2.3 Score of the Guessing Game

In addition to the adequacy of the images, the recognizability of the subjects from the images was assessed. Instead of trying to use another metric, a "guessing game" was introduced. Overall, it sent the images to a vision model, in case of this work, Llama-4-Scout-17B-16E-Instruct (Touvron et al. (2023), Meta (2025a), Meta (2025b)), together with a list of four potential subjects, asking the model to choose the portrayed subject. This allowed to check the identifiability of the images without a ground-truth image. For more data, there were multiple levels of the guessing game, and each one was repeated 6 times to reduce the impact of random guesses. For each, the correct and false guesses were counted, and the results were reported with their mean count and standard deviation.
Level 0, the easiest version, gave Llama the correct subject together with three other randomly chosen subjects from the same experiment. Level 1 had Llama choose between the true subject and three other subjects randomly chosen from within the same category as the original subject. These two levels were done for the Regular subjects, as illustrated by the Stable Diffusion model and Dall-E 2 (Ramesh et al. (2022)), as well as the Random subjects, illustrated by the same vision models. Additionally, the New Random subjects, as illustrated by the Stable Diffusion model, were also tested on both of these guessing game levels. Level 0 was also done for the Scale Test and its Reuse variant.
Levels 2 to 4 were only done with the Scale Test dataset. They handed Llama different counts of randomly selected tuples instead of the generated image. In ascending order of the levels 2, 3, and 4, the tuple counts were 5, 25, and 50 in one run, and 1, 2, and 3 in another to get more detailed results. All cases gave completely random options next to the true subject, like level 1, since the Scale Test dataset had not been split into categories due to its size.
Level 4
An example for the general layout of the game rules given to Llama, in addition to the image, can be seen in Fig. 3.1.

```
Let's play a guessing game! Look at
the image and guess which of these
4 subjects it is.  These are the
possible subjects: {guess_options}.
Respond with only the subject you
guessed and nothing more.
```

(a) General rule set for levels 0 and 1

```
Let's play a guessing game!  These
are some lines of a knowledge base
entry: {entry} Guess which of these
4 subjects it is.  These are the
possible subjects: {guess_options}.
Respond with only the subject you
guessed and nothing more.
```

(b) General rule set for levels 2 to 4

Figure 3.1: Generic prompts used in the guessing game levels

3.3 Generation Method

This section will go over the two main points of the used generation method: which model was used and why, and how the prompting was done. It shows how this work went about answering RQ 1 and RQ 2.

3.3.1 Model Choice

Whenever an AI is used, no matter the type of task, the model choice plays an essential role. Attached to it are major issues like finances and performance, which need to be optimized for each particular use case. In this thesis, multiple tasks had to be tackled. The most obvious one is the image generation. It needed a vision model that could continuously generate recognizable images of adequate quality for a large variety of subjects, some of which might not even exist. The most prominent options for vision models of that sort right now are Stable Diffusion models (Rombach et al. (2022)) or stacked, diffusion-like models like Dall-E 2 (Ramesh et al. (2022)).
With Dall-E 2 as a commercial model, come costs for each generation at $ 0.02 per generated image[1], as well as fast support and stable servers. For the tests, these expenses are within reason ($ 7.2 for one experiment generating all three knowledge sources for 120 subjects), but as this work aimed to create a scalable approach to illustrate an entire knowledge base, these costs would accumulate quickly. Even if only the same approximate image-entry ratio as Wikidata (around 5 % of entries illustrated) were intended, these costs would rise to $ 6,100 for 305,000 entries (around 5 % of GPTKB v1.5.1). Because of its broad usability, Dall-E 2 has filters for certain prompts, which will not be generated. Since these filters include things like famous people, monkeys, and apes, as well as specific word combinations and names, they were another point to be considered for the model choice. After all, every declined prompt needs a new prompt to be formulated, and the generation has to be attempted again. This has a large impact on time as well as the quality of the prompts, since paraphrasing can add subtle changes in meaning.
Another model available for this work was a Stable Diffusion model hosted by SCADS.AI, stable-diffusion-3.5-large-turbo. While the hosting process needs financing, the actual generation of images did not incur extra costs. Since the hosting is not attached to this work, however, these costs can be neglected for the consideration of a vision model for this thesis. Additionally, this more locally hosted model did not seem to have the kind of filters a publicly available, commercial model like Dall-E 2 has. This reduced some efforts in prompt generation, making it possible to avoid reformulating prompts altogether.

Manual review of the images generated by both models for the same subjects, as shown in Fig. 3.2, as well as comparison of CLIPScores, was used to finally decide on the model used for the rest of the thesis, as well as for the project of illustrating GPTKB in the future.

(a) Generated by Dall-E 2

(b) Generated by Stable Diffusion model

Figure 3.2: Images generated for the Apostolic Palace by Dall-E 2 and the Stable Diffusion model from a KB prompt

[1] as of the time of this thesis, April—September 2025

Furthermore, a model was needed to generate the prompts and another one to play the guessing game. The main prerequisite for the prompt generation model was a large context window, since full knowledge base entries were given to it at times. These could be as short as a single triple, but some reached over 20,000 tokens. Due to pricing and availability, as well as its large context window of 10 million tokens and big size at 17 billion activated parameters, this work used Llama-4-Scout-17B-16E-Instruct. It also used the same Llama model for the guessing game, since Llama-4-Scout-17B-16E-Instruct accepts images as input, and to not further complicate the matter by using a different model.

3.3.2 Prompting

The prompts to generate images from were generated via a SCADS.AI-hosted Llama-4-Scout-17B-16E-Instruct model. For the prompting, three different sources of knowledge for the prompts were tested originally. A prompt only leaving the vision model to fill in context that only consists of the general task and subject name; a prompt using the inherent knowledge of Llama as an LLM, made up of 3–5 sentences generated by Llama-4-Scout-17B-16E-Instruct; a prompt using the knowledge passed from the knowledge base entry generated by Llama when given instructions, the subject, and the GPTKB entry of the subject. From now on, these prompt types will be referred to as the sources of their knowledge: VLM prompt (VLMp), LLM prompt (LLMp), and KB prompt (KBp). For the KB prompt, Llama was explicitly tasked to use a style fitting the subject type. The general makeup of all three prompt types can be seen in Fig. 3.3.

```
Make this correct: An image of {subject}. Make sure
to respond with only the corrected word group and
nothing more. You are only allowed to add a or an if
necessary.
```

```
                         An image of the Apostolic Palace.
```

(a) VLMp prompt for the prompt generation through Llama

```
You are an AI artist. Create a prompt to generate a
representative image for this: {subject}. Do not be
too wordy. Keep the prompt to 3-5 sentences. Respond
with only the prompt and nothing more.
```

```
        Generate an image of the Apostolic Palace, the
        official residence of the Pope in Vatican City.
      Depict the palace's grand facade with its ornate
     Renaissance-style architecture and imposing stone
   walls. Include the palace's iconic courtyard and a
      glimpse of the Vatican's lush gardens. Use warm,
       golden lighting to evoke a sense of history and
                                          spirituality.
```

(b) LLMp prompt for the prompt generation through Llama

```
You are an AI artist and are supposed to create a
prompt for image generation. You are to create a
prompt for a representative image of this: {subject}.
The "gptkbp:instance_of" property usually states the
type of the subject. Make sure to get the right type
since some subjects might have ambiguous names.
Entry: {entry} If the entry has a "population"
property, you cannot under any circumstances use a
person type. Look for a non-person alternative with
the same name. The style should make sense for the
subject depicted. Real subjects like cities or
people should be depicted realistically. Respond
with only the prompt and nothing more.
```

```
A highly detailed and realistic depiction of the
Apostolic Palace, gptkbp:instance_of:palace, located
in Vatican City, showcasing its stunning Renaissance
architecture and ornate decorations, set against a
serene backdrop of Vatican Hill.
```

(c) KBp prompt for the prompt generation through Llama

Figure 3.3: All types of prompts given to Llama to generate an image generation prompt, as well as an example of the resulting prompt for the subject "Apostolic Palace" for each type (from the Regular dataset)

3.4 Scalability

This section will give an overview of the aspects of scalability that have been researched in this work, long-tail subjects, and saving resources by reusing images. The following will detail how RQ 3 and RQ 4 were set out to be answered.

3.4.1 Long-tail

A knowledge base includes a variety of entities, some well-known ones with large entries, and some lesser-known ones with very short entries. These lesser-known, short-entry subjects are often referred to as long-tail subjects, as they make up the long tail of the natural distribution of popularity across items (Fig. 3.4). As there is less information about these entities, the question stands whether their respective generated images are less representative than those of popular subjects.
To check this hypothesis, the subjects in the Regular, Random, and New Random datasets, as illustrated by the Stable Diffusion model, were split into the groups popular and long-tail. For both groups as well as each knowledge source of prompts, the CLIP and UQI scores were calculated, and their means and standard deviations were recorded. These scores could then be compared to investigate potential differences between the illustrations of popular and long-tail entities.

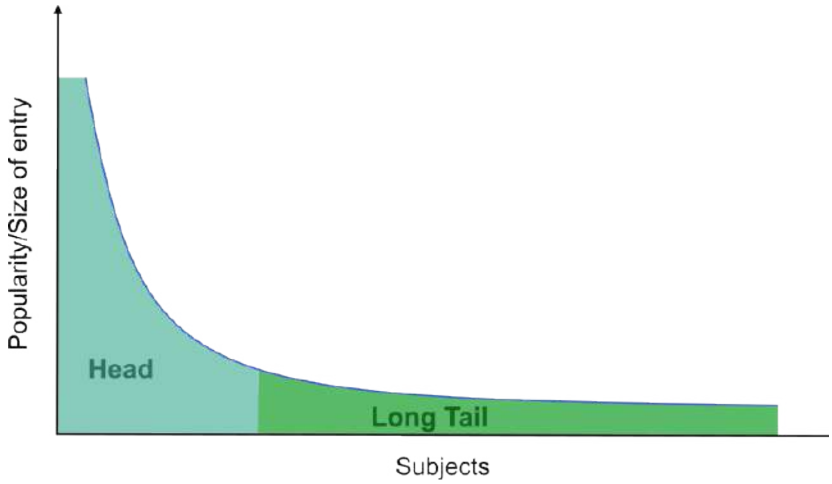


Figure 3.4: Graphic showing the distribution of subjects regarding their popularity

3.4.2 Reuse

Another aspect of scalability that needed to be explored was the idea of reuse. The question remains if 6.1 million images are necessary to sufficiently illustrate 6.1 million entries. If a preexisting image already sufficiently illustrates a new subject, aka leads to a sufficiently high CLIPScore for the subject, there is no need to generate a new image; instead, the existing one could be reused. During the latest, the large-scale generation, which was done in this work, generating images for over 100,000 entities of GPTKB, each image took around 4 to 5 seconds to be created, including entry extraction, prompt generation, and image generation. This meant the process for all 100,000 images took a bit over five whole days (around 125 h), meaning generation for the entire 6.1 million entries of GPTKB would take almost 318 days (around 7,625 h). If images could be reused, that could drastically lower the speed and potential generation cost. The reuse was investigated with the Regular and Scale Test datasets in different ways.
First, two different reuse cases were simulated for the Regular dataset. The reuse of companies for their products as well as the reuse of cities for their landmarks, and the opposite case. In both cases, the images to be reused were duplicated and renamed to replace the image of the subject they would be reused for. Then their CLIP and UQI scores were determined as usual and reported for just the affected categories of subjects, as well as overall, to see if the mean scores were heavily affected by the reuse. This experiment was meant to determine if reuse was a worthwhile option to look into further. An example of which image was reused in the two cases can be seen in Fig. 3.5.

(a) "Vatican City" used for "Vatican City" and its landmark, "Apostolic Palace"

(b) "Apostolic Palace" used for "Apostolic Palace" and its city, "Vatican City"

Figure 3.5: The two cases of reuse for the City-Landmark pair Vatican City and Apostolic Palace, each shown with the image generated from the KB prompt

The reuse experiment with the Scale Test dataset was done more generally, to determine the actual usability of reuse on a large-scale image generation and how much reuse was feasible. A subset of 500 subjects was randomly chosen from all 5000 subjects in the dataset. Each of these 500 subjects had its most similar subject from the other 4999 subjects found. For this similarity check, the entries of the subjects were compared through paraphrase mining with the mixedbread-ai/mxbai-embed-large-v1 model (Lee et al. (2024), Li and Li (2024)) with the "sentence_transformers" (Reimers and Gurevych (2019)) library for Python. Then, a CLIPScore evaluation was done, simulating more and more reuse. On the first run, all the actual images generated for the subset of 500 subjects were used and assessed. Then, the 10 % of subjects with the highest similarity scores to another subject had their images replaced by that of the most similar subject, and all 500 images were assessed via CLIPScore. This continued until all subjects were being evaluated with a replacement image. From the resulting curve, assumptions about the best percentage of reuse could be made.

3.5 Utility and Bias

In this section, the methodology used to answer RQ 5 and RQ 6 will be shown. It includes finding a generated image's value in comparison to the tuples of an entry, as well as researching potential biases in the generated data.

3.5.1 Image Worth

To determine the worth of a generated image added to a knowledge base's entry, this work used the aforementioned guessing game. Specifically, a comparison of how easily an image was recognized as a subject with how easily different counts of tuples were recognized as a subject was done. This was done by using the same 500-subject subset of the Scale Test dataset used in the percentage-wise reuse check with their original images. Six different counts of tuples were tested: 1, 2, 3, 5, 25, and 50.

By comparing the false and correct guesses when using images and when using tuples, the information value of the two regarding recognizability can be put in relation. The determined worth could be understood as the count of tuples that an image could replace.

3.5.2 Biases

While checking for biases is always an important step in scientific research, it is even more common with AI, especially GenAI (Zhou et al. (2024), Wei et al. (2025)). To study potential biases in the data generated in this thesis, 500 images from the final 100,000 generated images were chosen. They were only chosen if Llama-4-Scout-17B-16E-Instruct determined that the image had a central person whose traits, ethnicity, age, and gender could be guessed. If such a person was not in the image, a new one was chosen until the desired number of 500 images was obtained. The chosen images were then given to Llama 4, one by one, and Llama was asked to identify the central person's ethnicity, age, and gender. The general prompt given to Llama, in addition to the image, as well as an example image and its guessed traits, can be seen in Fig. 3.6.

```
Look at the given image and
guess certain traits of the central
person in the image. Your
answer should have this format:
[Ethnicity];[Age];[Gender] Ethnicity
and Gender have to be a single word
each. Be sure not to use potentially
racist words for Ethnicity. Age has
to be a single integer. Do not write
the [] and only return the filled in
traits as shown before. Make sure you
give all answers for the image given!
```

(a) The generic prompt for Llama

(b) Generated image of "J.P. Morgan" with the guessed traits: Caucasian, 70, Male

Figure 3.6: Generic prompt for trait-guessing as well as an example image given to Llama 4 and its guessed traits

The traits Llama 4 guessed were then partially compared to the overall distribution of that trait in GPTKB, since GPTKB is what should be accurately portrayed by the images. It was not possible to do so for the age, since only very few GPTKB entries have an age field. Some have birth or death years or both, but the age at which a person died is not necessarily the age associated with the person. For gender and ethnicity, approximations could be gathered through queries.
For the gender, only the counts of "female" and "male" were considered as the ground-truth of GPTKB. There were too many synonymous options to add all of them to the two main categories, and some entries might have multiple of them as well.
For the ethnicity, it was a bit more complicated. Since ethnicities can be referred to in even more different terms, GPTKB counted over 4,800 different ethnicities in its database. Instead of trying to organize all of them, it was decided to gather the ten most common nationalities represented in GPTKB. The official ethnicity distributions of those nations were then used to estimate a ground-truth distribution of ethnicities in GPTKB. Since the ten most common nationalities included both "British" and "English", their occurrences were summed up, and the 11th most common nationality, "Russian", was used as the 10th nationality. The approximate distributions of ethnicities were inferred from Wikipedia (2025). For better comparability, the ethnicities on that list were fit into the more general ethnicities that had been guessed by Llama 4. The query and its results can be found in Fig. 3.7.

```
PREFIX gptkb: <https://gptkb.org/entity/>
PREFIX gptkbp: <https://gptkb.org/prop/>

SELECT ?o (COUNT(*) AS ?ofreq)
WHERE {
?s gptkbp:nationality ?o.
}
GROUP BY ?o
ORDER BY DESC(?ofreq)
LIMIT 11
```

(a) GPTKB query

Nationality	Frequency
American	373671
British	133951
French	48306
German	45389
Indian	45092
Canadian	41608
Australian	30981
Japanese	30452
Italian	29333
English	20194
Russian	17659

(b) GPTKB query result

Figure 3.7: Query used to gather the most common ethnicities in GPTKB, as well as the query result

4 Results

This chapter will show the results of the experiments done in this thesis. The focus will be solely on the results, not on any implications made by them.

4.1 Generation Method

In this section, all results regarding the generation method will be detailed. This includes the results of the experiments done to decide on the vision model used, as well as the decision for the type of prompting to be used.

4.1.1 Used Model

As previously described in section 3.3.1, there were two main factors studied besides costs regarding the model choice: the comparison of UQI, CLIP, and guessing game scores, and a manual review of the generated images for the same subjects. The more objective of the two was the comparison of CLIP, UQI, and guessing game scores. Tab. 4.1 shows a list of CLIP and UQI scores for Dall-E 2 and the Stable Diffusion model. Since the New Random dataset was only used with the Stable Diffusion model, no data for Dall-E 2 regarding that dataset is available.

The UQI scores were very similar across all experiments and models, in both mean values and standard deviations. They kept around a mean of 0.50 and a standard deviation of 0.20 throughout.
The CLIPScore mean values and standard deviation varied substantially more across experiments, knowledge sources, and models. Since the standard deviations of both models were quite similar, at 2.4 and 2.9 for Dall-E 2, and 3.0, 3.0, and 2.9 for the Stable Diffusion model, the CLIPScores can be directly compared. Dall-E 2 reached a lowest CLIPScore average across the knowledge sources of 90.2 for the Random dataset, and a highest one of 91.5 for the Regular dataset, making it a range in score of 1.3. In comparison, the Stable Diffusion model had a lower range in means of just 0.6 and higher absolute values on all accounts. Its lowest value for a CLIPScore average was a score of approximately 91.3 for the New Random and the Random dataset. That means that its worst performance had a difference to Dall-E 2's best performance of only 0.2, while the difference of Dall-E 2's worst performance

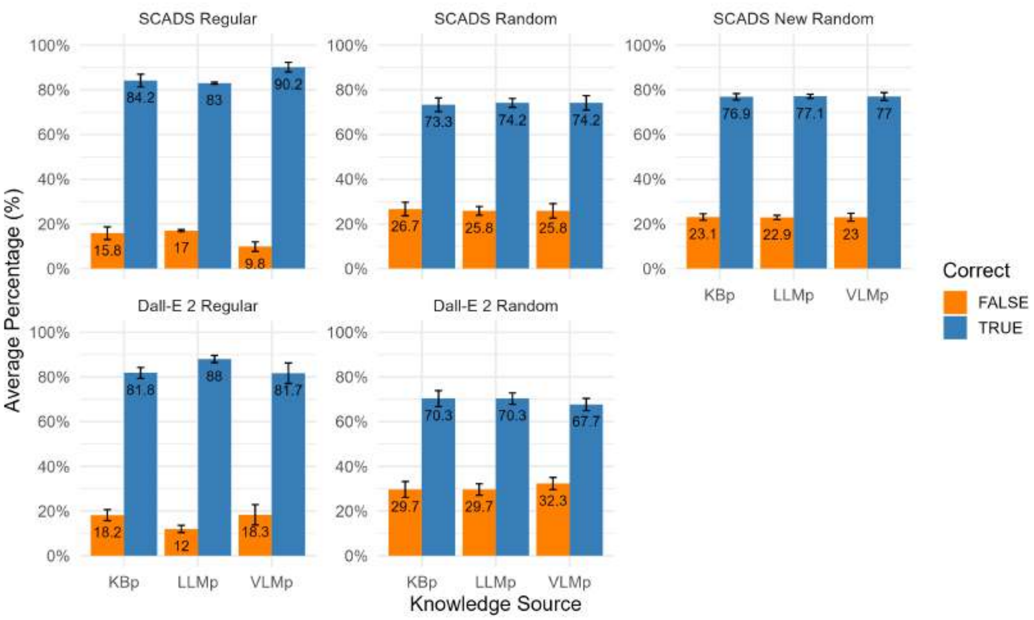
to Stable Diffusion's best was 1.7. The Stable Diffusion model's best performance was also for the Regular dataset at an average of 91.9.

Dataset	Avg. CLIP $\pm$ SD	UQI Mean $\pm$ SD
Dall-E 2 Regular	91.5 ± 2.4	0.48 ± 0.24
Dall-E 2 Random	90.2 ± 2.9	0.50 ± 0.19
SCADS Regular	91.9 ± 3.0	0.50 ± 0.24
SCADS Random	91.3 ± 3.0	0.48 ± 0.22
SCADS New Random	91.3 ± 2.9	0.58 ± 0.16

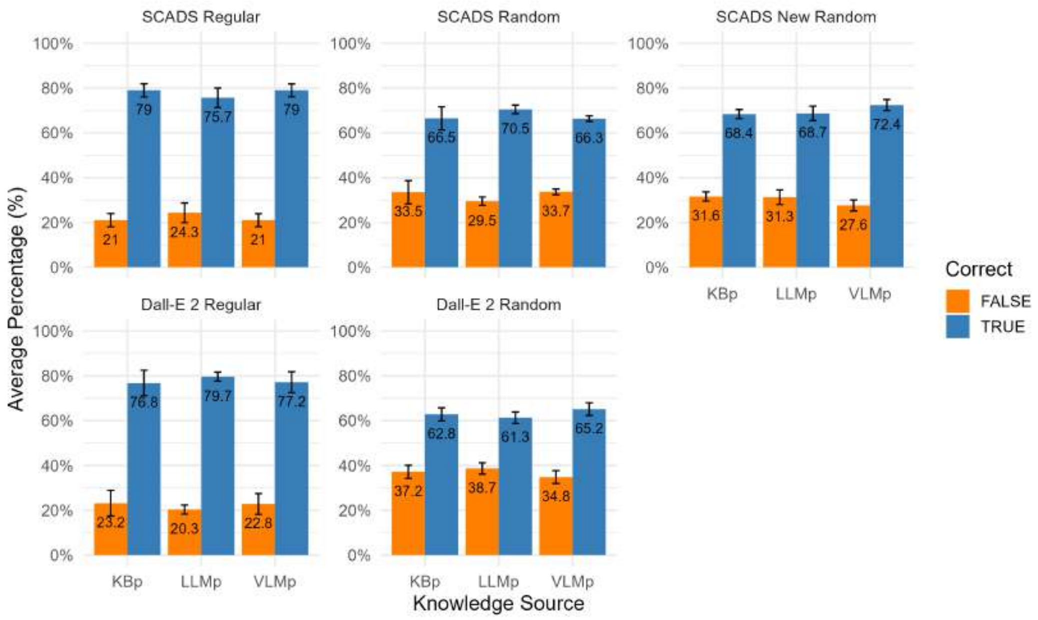
Table 4.1: CLIP and UQI Scores by dataset and knowledge source (KS) for Dall-E 2 and the Stable Diffusion model, as well as the average (avg.) CLIPScore for all knowledge sources each

For the guessing game scores, both the mean percentage of correct and incorrect guesses and the standard deviations of the prediction correctness are reported. Fig. 4.1 shows a visualization of the prediction accuracy across 6 tests each, for the same model-dataset combinations as Tab. 4.1. The results for the guessing game level 0, in which the three other guess options were chosen from the entire pool of subjects of the particular dataset, are visualized in Fig. 4.1a. The results for level 1, in which the three other guess options were chosen from the same category as the true subject, within the particular dataset, can be seen in Fig. 4.1b.

The guessing game scores for level 0 clearly show that, no matter the model, the Regular dataset gets more correct guesses in comparison to the Random or New Random datasets. For the Stable Diffusion model, the averaged percentages over the three knowledge sources are 85.8 % for Regular, 73.9 % for Random, and 77.0 % for New Random. For Dall-E 2, they are 83.8 % for Regular, and 69.4 % for Random. Just like with the CLIPScores, the Stable Diffusion model performs better in all experiments and for all knowledge sources on average. Additionally, the only individual value that is better for Dall-E 2 in comparison to the Stable Diffusion model is the LLM prompt for the Regular dataset, with 88 % correct guesses for Dall-E 2 and 83 % correct guesses for the Stable Diffusion model.



(a) Guessing game level 0



(b) Guessing game level 1

Figure 4.1: Prediction accuracy by knowledge source and dataset for guessing games level 0 and level 1 (with SD error bars across 6 tests)

For level 1, the guessing game scores look slightly different, and the accuracy is lower than in level 0 for both models. For the Regular dataset, both Dall-E 2 and the Stable Diffusion model have an average accuracy of 77.9 %. Both the Random and New Random values for the Stable Diffusion model, at an average correct percentage of 67.8 % and 69.8 %, respectively, are higher than Dall-E 2's average percentage for Random, at only 63.1 %. Just like with level 0, all individual values for each of the knowledge sources within the different datasets, except for the LLM prompt of the Regular dataset, are higher for the Stable Diffusion model.

Model and Dataset	KS	Level 0	Level 1	Level 2	Level 3	Level 4
stable diff. Regular	KBp	2.8	3.0	-	-	-
	LLMp	0.5	4.4	-	-	-
	VLMp	2.2	2.9	-	-	-
stable diff. Random	KBp	3.1	5.2	-	-	-
	LLMp	2.0	1.9	-	-	-
	VLMp	3.2	1.3	-	-	-
stable diff. New Random	KBp	1.4	2.0	-	-	-
	LLMp	1.0	3.3	-	-	-
	VLMp	1.8	2.5	-	-	-
Dall-E 2 Regular	KBp	2.5	5.7	-	-	-
	LLMp	1.6	2.0	-	-	-
	VLMp	4.6	4.7	-	-	-
Dall-E 2 Random	KBp	3.6	2.9	-	-	-
	LLMp	2.5	2.5	-	-	-
	VLMp	2.7	2.9	-	-	-
Scale Test	KBp	0.6	-	2.0	1.6	1.1
Scale Test Reuse	KBp	1.1	-	-	-	-

Table 4.2: Standard deviation for count percentages in guessing game levels 0–4. 0: Random guess options from all possible subjects; 1: Random guess options from the same category; 2: No image, 1 tuple of entry, random options; 3: No image, 2 tuples of entry, random options; 4: No image, 3 tuples of entry, random options

To consider randomness in the guesses of Llama-4, the guessing games have each been performed six times, and the standard deviations for guessing correctly or incorrectly have been determined. All standard deviations, in addition to those for the guessing games done with the Scale Test dataset, with the Stable Diffusion model, for the image worth determination can be found in Tab. 4.2. Two trends can be seen in the data. Firstly, the prediction accuracy in level 1 tends to vary more than in level 0. Out of 15 experiments with both levels, 12 showed a higher standard deviation in level 1, with some having an increase of over 200 %, and Stable Diffusion's Regular LLM prompt even increasing by over 700 % from level 0 to level 1. Secondly, in all cases but the stable diffusion experiments for level 1, the LLM prompt had the lowest standard deviation.

4.1.2 Prompts

Together with choosing a model for future large-scale generation, a prompting approach had to be chosen. In Tab. 4.3, the CLIP and UQI scores for the individual knowledge sources in the experiments comparing the Dall-E 2 and the Stable Diffusion model can be seen.

Concerning the UQI scores, the results stayed the same as for the model choice experiments. The means were all around 0.50, and the standard deviations were around 0.20.

Again, the CLIPScores showed clearer distinctions. Next to the fact that all individual values for the Stable Diffusion model were higher than the corresponding values for the Dall-E 2 model, trends regarding the knowledge sources can be observed. For both Dall-E 2 experiments and the corresponding SCADS experiments, the LLM prompt shows the highest CLIPScores. For the New Random dataset, however, which was generated through the Stable Diffusion model and has 200 subjects instead of the 120 of the others, the KB prompt had the best performance. While the KB prompt showed the worst performance for Regular and Random for the Stable Diffusion model, it was the second-best for both Dall-E 2 experiments.

Model and Dataset	KS	CLIP Mean $\pm$ SD	UQI Mean $\pm$ SD
stable diff. Regular	KBp	91.7 $\pm$ 3.0	0.56 $\pm$ 0.23
	LLMp	92.1 $\pm$ 2.7	0.44 $\pm$ 0.25
	VLMp	92.0 $\pm$ 3.2	0.51 $\pm$ 0.24
stable diff. Random	KBp	91.0 $\pm$ 3.1	0.52 $\pm$ 0.23
	LLMp	91.6 $\pm$ 2.9	0.44 $\pm$ 0.22
	VLMp	91.3 $\pm$ 3.0	0.48 $\pm$ 0.22
stable diff. New Random	KBp	91.8 $\pm$ 2.9	0.57 $\pm$ 0.16
	LLMp	91.0 $\pm$ 2.8	0.57 $\pm$ 0.15
	VLMp	91.0 $\pm$ 2.9	0.59 $\pm$ 0.17
Dall-E 2 Regular	KBp	91.4 $\pm$ 2.5	0.46 $\pm$ 0.24
	LLMp	91.5 $\pm$ 2.8	0.47 $\pm$ 0.24
	VLMp	91.5 $\pm$ 3.2	0.51 $\pm$ 0.25
Dall-E 2 Random	KBp	90.2 $\pm$ 2.8	0.49 $\pm$ 0.19
	LLMp	90.4 $\pm$ 2.8	0.47 $\pm$ 0.18
	VLMp	89.9 $\pm$ 3.1	0.55 $\pm$ 0.21

Table 4.3: CLIP and UQI Scores by dataset and knowledge source (KS) for Dall-E 2 and the Stable Diffusion model

4.2 Scalability

This section will conclude the results of the test most directly regarding the scalability of the concept, the effect of long-tail subjects on the scores, and the influence of reuse on the image adequacy.

4.2.1 Long Tail

To gauge the influence of subject popularity on the image adequacy as portrayed by the CLIPScores, experiments were conducted. The results can be found in Tab. 4.4, showing both CLIP and UQI scores for images generated by Dall-E 2.

The average of the UQI scores overall is still approximately 0.50, but there are still some trends visible. For all popular items, the KB prompt has the best UQI score, and, except for the Random dataset, it is higher for the popular items than for the long-tail items. The LLM prompt is the only one that has a consistently lower UQI score for the popular subjects than for the long-tail subjects. For the VLM prompt, the better score varies between popular and long-tail subjects. The Random dataset, in contrast to the Regular dataset and the Reuse cases derived from the Regular dataset, shows better UQI scores for the long-tail items for all three knowledge sources. The KB prompt has the highest score for both popularities.

Dataset	Popularity	KS	CLIP Mean $\pm$ SD	UQI Mean $\pm$ SD
Regular	Popular	KBp	91.6 ± 2.6	0.60 ± 0.19
		LLMp	91.8 ± 2.4	0.55 ± 0.16
		VLMp	93.4 ± 2.5	0.56 ± 0.16
	Long-tail	KBp	91.4 ± 2.5	0.54 ± 0.21
		LLMp	91.9 ± 3.0	0.56 ± 0.22
		VLMp	91.9 ± 3.5	0.52 ± 0.20
Reuse-CC	Popular	KBp	90.1 ± 2.1	0.60 ± 0.16
		LLMp	90.2 ± 2.1	0.56 ± 0.17
		VLMp	90.6 ± 2.0	0.51 ± 0.21
	Long-tail	KBp	90.2 ± 2.8	0.56 ± 0.22
		LLMp	89.7 ± 3.0	0.49 ± 0.19
		VLMp	90.8 ± 1.5	0.52 ± 0.21
Reuse-PL	Popular	KBp	91.4 ± 3.0	0.59 ± 0.18
		LLMp	91.8 ± 2.5	0.55 ± 0.16
		VLMp	93.1 ± 3.1	0.55 ± 0.17
	Long-tail	KBp	91.5 ± 2.0	0.56 ± 0.21
		LLMp	91.8 ± 3.1	0.58 ± 0.22
		VLMp	92.0 ± 3.7	0.53 ± 0.20
Random	Popular	KBp	91.0 ± 3.6	0.47 ± 0.15
		LLMp	91.6 ± 2.3	0.39 ± 0.20
		VLMp	91.4 ± 2.9	0.32 ± 0.15
	Long-tail	KBp	90.2 ± 2.7	0.56 ± 0.21
		LLMp	90.3 ± 2.7	0.49 ± 0.21
		VLMp	89.7 ± 3.0	0.52 ± 0.20

Table 4.4: CLIP and UQI Scores for Regular, Reuse-CC (Cities and Companies use reused images), Reuse-PL (Landmarks and Products use reused images), and Random by Dataset and Knowledge Source (KS) for Dall-E 2

The CLIPScores show some consistency as well. For all cases but the Reuse cases, the popular items have slightly higher CLIPScores. The largest difference in score can be seen for the Random dataset, at a mean decrease of CLIPScores of about 1.3 from popular to long-tail subjects. In comparison, the average decrease for the Regular dataset is only 0.5, and the average decreases for the Reuse cases are even lower at 0.1 for Reuse-CC and at 0.3 for Reuse-PL. The Regular dataset shows the largest increase in standard deviation between popular (2.5) and long-tail (3.0) subjects, when compared to Random (2.9 to 2.8), reusing images for city and company images (2.1 to 2.4), and Reuse of product and landmark images (2.9 to 2.9).

Group	Popularity	KS	CLIP Mean $\pm$ SD	UQI Mean $\pm$ SD
Regular	Popular	KBp	92.5 ± 2.9	0.62 ± 0.13
		LLMp	92.9 ± 3.0	0.52 ± 0.18
		VLMp	93.9 ± 3.0	0.64 ± 0.17
	Long-tail	KBp	92.3 ± 1.6	0.57 ± 0.21
		LLMp	91.6 ± 2.0	0.53 ± 0.23
		VLMp	92.0 ± 3.2	0.61 ± 0.23
Random	Popular	KBp	93.0 ± 2.8	0.43 ± 0.21
		LLMp	93.1 ± 2.6	0.33 ± 0.21
		VLMp	93.2 ± 2.5	0.57 ± 0.18
	Long-tail	KBp	90.7 ± 3.1	0.51 ± 0.23
		LLMp	91.5 ± 2.8	0.48 ± 0.22
		VLMp	91.0 ± 3.0	0.53 ± 0.25
New Random	Popular	KBp	93.5 ± 2.6	0.57 ± 0.17
		LLMp	92.9 ± 2.7	0.55 ± 0.18
		VLMp	91.6 ± 2.7	0.55 ± 0.16
	Long-tail	KBp	91.6 ± 2.9	0.59 ± 0.16
		LLMp	90.9 ± 2.8	0.58 ± 0.15
		VLMp	91.0 ± 3.0	0.57 ± 0.17

Table 4.5: CLIP and UQI Scores for Regular and Random Datasets by Popularity and Knowledge Source (KS) for Stable Diffusion model

For the Stable Diffusion model, the values look a bit different, as seen in Tab. 4.5.
The UQI scores are about the same as the ones for Dall-E 2, averaging at around 0.50, and do not seem to exhibit consistent differences between popular and long-tail subjects. Their standard deviations also look very similar to the ones for Dall-E 2, still averaging around the 0.20 mark.
The CLIPScores show differences between the popular and long-tail subjects. All of them have higher scores for the popular entities. Additionally, the KB prompt performs the best for the long-tail subjects at an average distance to the other prompt types of 0.65 for New Random, -0.55 for Random, and 0.50 for Regular. Its average distance overall is therefore 0.2, compared to -0.1 average distance from the other prompts for both LLMp and VLMp. Concerning the popular subjects, the trend is in favor of the VLM prompt, with average differences to the other knowledge sources of -1.1 for New Random, 0.15 for Random, and 1.2 for Regular. This concludes in an overall average of 0.08, compared to 0.02 for LLMp and 0.07 for KBp.

4.2.2 Reuse

The first experiment regarding reuse was to look at the averages over just the two categories affected by the reuse, instead of the entire dataset. Tab. 4.6 shows the UQI (Tab. 4.6a) and CLIP (Tab. 4.6b) scores from this experiment for just the affected categories. The values in the column "Reuse Cases" depend on the type of Reuse, reusing images for the labeled category's subjects. In that column, all values labeled "Cities" or "Companies" stem from the use of reused images for city or company images, and all values labeled "Landmarks" or "Products" stem from the use of reused images for landmark or product images.

		(a) UQI Scores		(b) CLIP Scores	
		Regular	Reuse Cases	Regular	Reuse Cases
Group	KS	Mean $\pm$ SD	Mean $\pm$ SD	Mean $\pm$ SD	Mean $\pm$ SD
All	KBp	0.47 $\pm$ 0.26		91.6 $\pm$ 2.9	
All	LLMp	0.47 $\pm$ 0.23		91.5 $\pm$ 2.8	
All	VLMp	0.51 $\pm$ 0.25		91.5 $\pm$ 3.2	
Cities	KBp	0.65 $\pm$ 0.11	0.68 $\pm$ 0.01	92.3 $\pm$ 5.6	90.7 $\pm$ 2.7
Cities	LLMp	0.64 $\pm$ 0.12	0.58 $\pm$ 0.02	93.4 $\pm$ 6.3	90.6 $\pm$ 3.2
Cities	VLMp	0.59 $\pm$ 0.11	0.55 $\pm$ 0.05	93.9 $\pm$ 11.3	90.8 $\pm$ 2.1
Companies	KBp	0.31 $\pm$ 0.33	0.34 $\pm$ 0.33	91.0 $\pm$ 2.7	89.5 $\pm$ 2.0
Companies	LLMp	0.25 $\pm$ 0.31	0.25 $\pm$ 0.30	89.8 $\pm$ 2.6	88.9 $\pm$ 1.9
Companies	VLMp	0.34 $\pm$ 0.36	0.25 $\pm$ 0.30	89.3 $\pm$ 2.6	89.3 $\pm$ 1.4
Landmarks	KBp	0.67 $\pm$ 0.11	0.62 $\pm$ 0.10	92.9 $\pm$ 2.8	91.1 $\pm$ 2.9
Landmarks	LLMp	0.58 $\pm$ 0.13	0.65 $\pm$ 0.11	92.7 $\pm$ 2.7	91.5 $\pm$ 2.7
Landmarks	VLMp	0.51 $\pm$ 0.22	0.57 $\pm$ 0.11	93.5 $\pm$ 2.4	91.9 $\pm$ 3.0
Products	KBp	0.51 $\pm$ 0.25	0.56 $\pm$ 0.26	92.4 $\pm$ 3.2	89.2 $\pm$ 1.9
Products	LLMp	0.43 $\pm$ 0.26	0.39 $\pm$ 0.28	91.7 $\pm$ 2.3	89.5 $\pm$ 1.5
Products	VLMp	0.46 $\pm$ 0.31	0.49 $\pm$ 0.29	92.5 $\pm$ 3.1	89.0 $\pm$ 2.6

(a) UQI Scores (b) CLIP Scores

Table 4.6: CLIP and UQI Scores by Knowledge Source (KS); All Subjects and Filtered for Regular and both Reuse cases (Images of Cities and Companies reused for Landmarks and Products and opposite; labeled after the category that had their images exchanged for a reused image) for Dall-E 2

Standard deviations for UQI regarding this experiment were at 0.20, and while the average UQI score was still about 0.50, the company images seem to have a slightly decreased UQI score with and without reuse. The average for just the company images was at 0.28 during the Reuse, and at 0.30 for just the Regular dataset.
The CLIPScores, once again, show clearer trends. All values for the Reuse cases are lower than for just the Regular dataset. The scores for the company images were the lowest, for both Regular and Reuse, and had the smallest average decrease in score between Regular and Reuse of just 0.47. In contrast, cities showed a decrease of 2.50, landmarks of 1.53, and products of 2.97. The VLM prompts showed the best scores for the Regular dataset, but none stood out in particular for the Reuse cases.

The second experiment was to compare the recognizability of subjects. Images specifically created for that subject had their guessing game results compared to reused images of the subjects that had the semantically most similar GPTKB entries to those original subject's scores. For that, a 500-subject subset of the Scale Test was used. The guessing game results can be seen in Fig. 4.2.

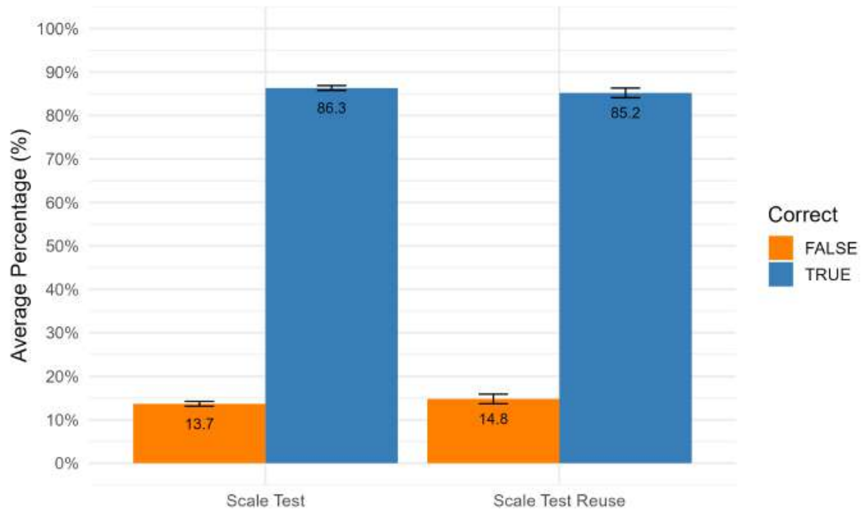


Figure 4.2: Prediction accuracy by group for guessing game level 0 for the 500-subject Scale Test subset (with SD error bars across 6 tests)

While the Reuse case shows a slightly worse accuracy with a higher standard deviation, the scores overall are not much different, with an average correct guess percentage of 86.3 % for the generated and 85.2 % for the reused images.

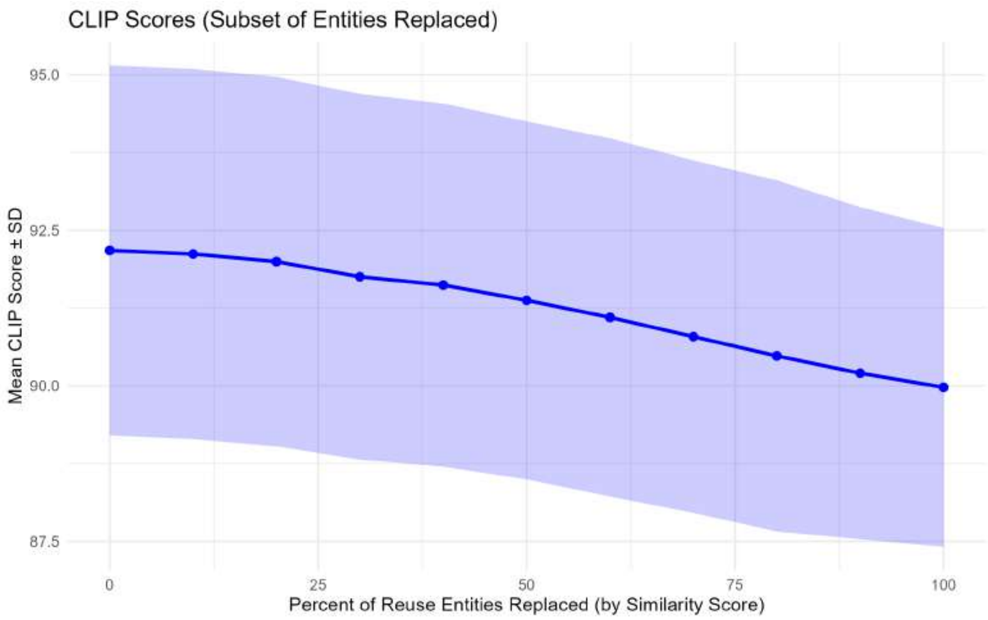


Figure 4.3: Average CLIPScores for the 500-subject Scale Test subset by the percentage of reused images; Reuse in order of descending similarity scores

The third and last experiment regarding Reuse was to generate a curve of reuse percentage to mean CLIPScore. For this, a 500-subject subset from the Scale Test dataset, generated by the Stable Diffusion model, was used. Fig. 4.3 shows the measured curve as well as the standard deviation, marking an error range for the CLIPScore values. The difference in CLIPScore between 0 % of entities reusing their most similar entity's image and 100 % is about 2.2. The curve also seems to decrease more slowly in CLIPScore toward the extremes, and more quickly between 25 % and 75 %, though there is a sudden drop from 20 % reused images to 30 % reused images. The measures CLIPScore range across all reuse percentages is approximately 92.2–90.0.

4.3 Utility and Bias

In this section, all results regarding the actual utility of the images for their intended purpose of illustrating entities of a knowledge base are presented. This includes the worth of an image in comparison to the triples in an entry, as well as potential biases.

4.3.1 Image Worth

To have a point to compare an image's worth to a triple's, the guessing game was slightly modified to give Llama-4 tuples from a subject's entry instead of images. This was done for the 500-subject subset of the Scale Test dataset and two different sets of tuple counts.

The first tuple counts were 5, 25, and 50 tuples. If a tuple count was greater than the actual count of tuples for a subject, this meant that all available tuples were used, and it simply did not reach the actual count. As seen in Fig. 4.4, the results were very similar across the three different tuple counts. All of them had over 99 % correct guesses on average.

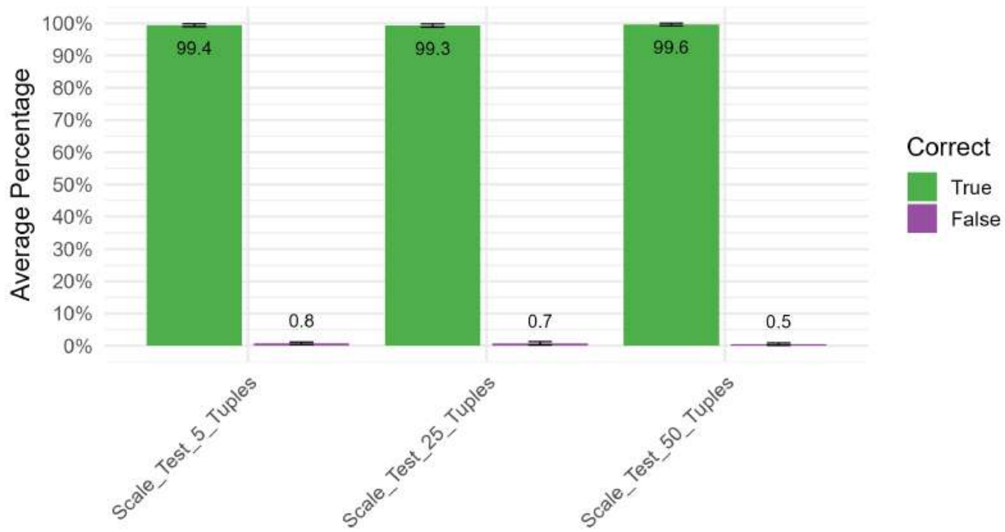


Figure 4.4: Prediction accuracy by group for guessing game levels 2, 3, and 4 (5, 25, and 50 tuples) for the Scale Test

Since the average count of triples per subject on GPTKB is 19.70[1], the two higher counts of tuples were often not filled out. Because of that, and to get more detailed results, the second set of tuple counts was used. It used the tuple counts 1, 2, and 3. The guessing game results for that can be seen in Fig. 4.5. 3 tuples still caused a very comparable result to the three higher counts of tuples at an average of 98.7 % correct guesses. 2 tuples caused a lower, but still comparable result with 97.1 % of guesses being correct. The lowest tuple count, 1, showed significantly different results. It had a drastically lower correct prediction average at just 93.0 % correct guesses on average over its six conducted guessing games.

[1] Query as of 03.10.2025

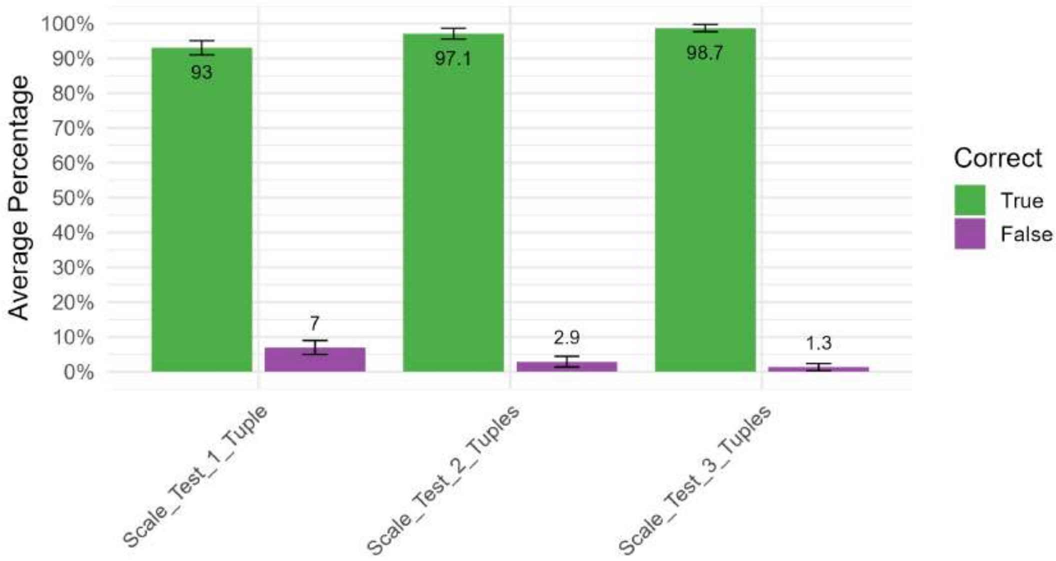


Figure 4.5: Prediction accuracy by group for guessing game levels 2, 3, and 4 (1, 2, and 3 tuples) for the Scale Test

4.3.2 Biases

Three different traits, ethnicity, age, and gender, were checked for biases within a 500-image, persons-only subset of the final Large Scale dataset.
The distribution of guessed ethnicities by Llama-4 in this subset, as well as the approximate distribution of ethnicities across GPTKB[2], can be seen in Fig. 4.6.

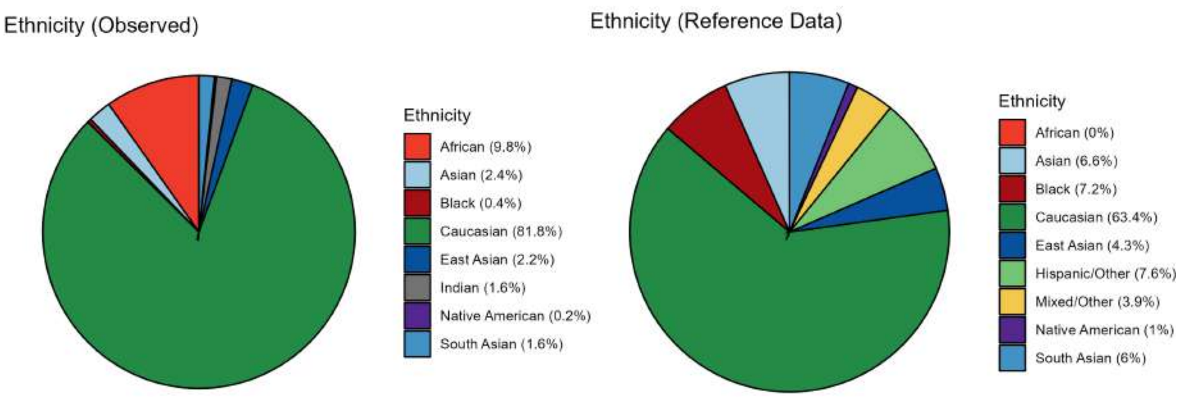


Figure 4.6: Pie charts for the guessed ethnicities by Llama-4 in the 500-image, persons-only subset of the Large Scale dataset and the approximate distribution of ethnicities across GPTKB (inferred from nationalities)

In both cases, "Caucasian" makes up the majority, though even if "Hispanic/Other" in the reference data from GPTKB is added to the "Caucasian" count, it only reaches 71 %, in comparison to the 81.8 % in the observed data. Other clear differences are the frequencies of "Black" and "African". While the observed data shows 9.8 % "African" and just 0.4 % "Black", the reference data has no "African" at all (0 %) but shows 7.2 % "Black".

[2]As inferred from the nationalities, as explained in 3.5.2

Furthermore, Llama-4 guessed four different ethnicities across the Asian continent: "Asian" (2.4 %), "East Asian" (2.2 %), "Indian" (1.6 %), and "South Asian" (1.6 %), with an overall presence of 7.8 %. The reference data, however, only distinguishes three different ethnicities across the Asian continent: "Asian" (6.6 %), "East Asian" (4.3 %), and "South Asian" (6 %), with an overall presence of 16.9 %. "Native American" is present in both charts, though the observed frequency of 0.2 % is only one-fifth of the reference frequency of 1 %. The last thing to note is the lack of "Mixed/Other" in the observed data, while it makes up 3.9 % of the reference data.

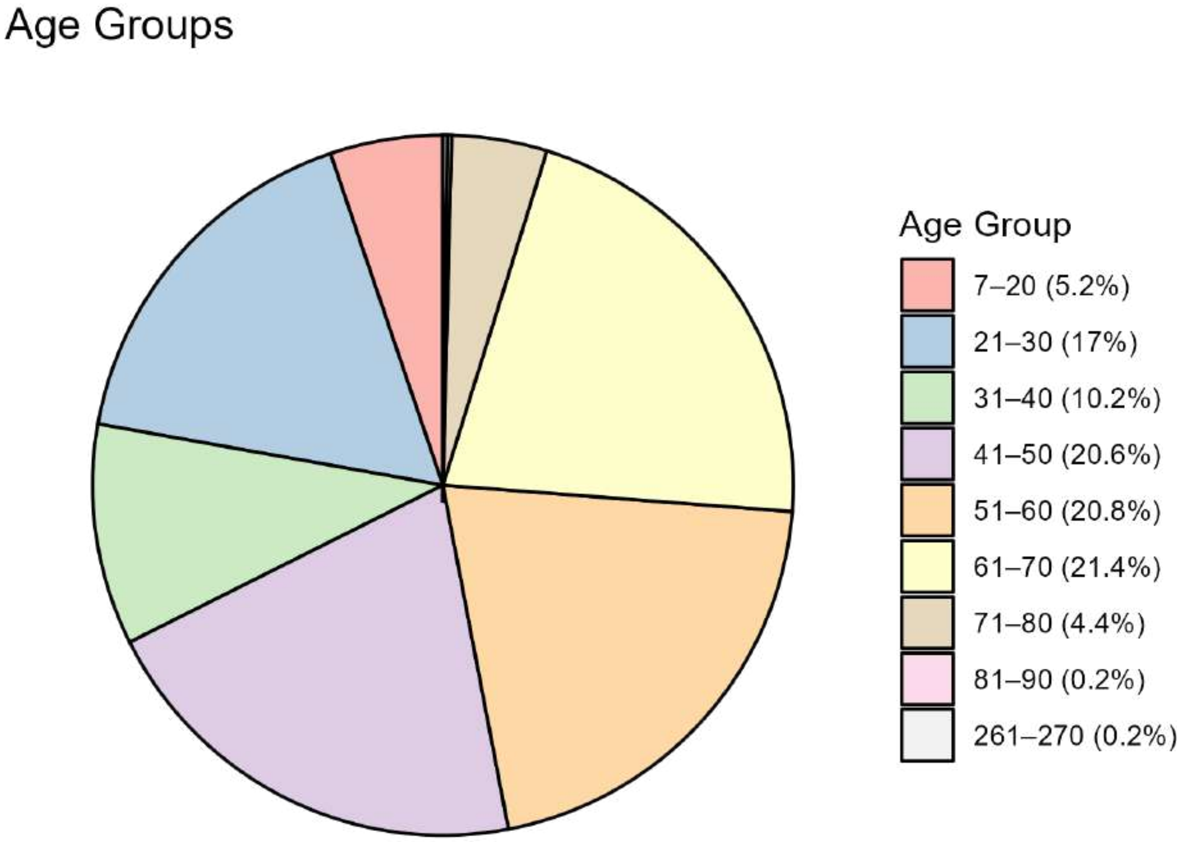


Figure 4.7: Pie chart of the observed ages across the 500-image, persons-only subset of the Large Scale dataset (guessed by Llama-4)

For the age bias check, no reference data could be inferred from GPTKB, so Fig. 4.7 shows only the observed data. The lowest age guessed by Llama-4 was 7 years, while the highest was 270 years. The majority of ages lie between 41 and 70, making up over two-thirds of the data (62.8 %). Each of the three age groups in that range (41—50, 51–60, and 61–70) holds around the same percentage, at 20.6 %, 20.8 %, and 21.4 % in ascending order of ages. The next largest age group is 21–30, making up 17 % of the data, followed by 31–40 by some distance at 10.2 %. At only 5.2 % and 4.4 %, the youngest age group (7–20) and 71–80 have about the same frequency. They are only undercut by the two oldest age groups, 81–90 and 261–270, at 0.2 % from just a single person each.

The comparison between the observed genders and the frequency within GPTKB is the most drastic. For ease of comparison and due to limitations of querying, only the distribution of the binary genders "male" and "female" was studied. The results can be seen in Fig. 4.8.

While GPTKB has close to a 50:50 split between the genders, with "female" slightly in the lead at 53.8 %, Llama-4 guessed almost three-fourths of the central persons in the generated images as "male", leaving "female" at just 26 %.

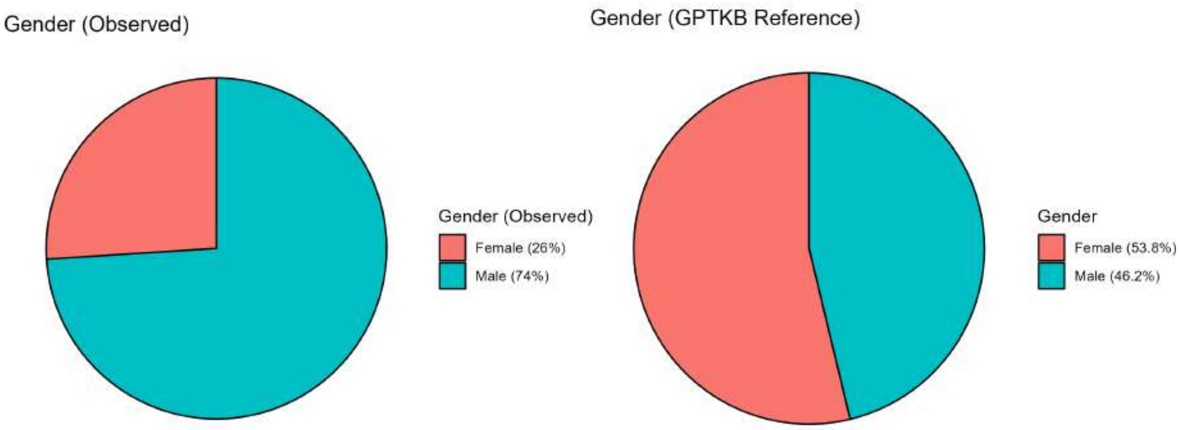


Figure 4.8: Pie charts for the assumed gender by Llama-4 in the 500-image, persons-only subset of the Large Scale dataset and the approximate distribution of genders across GPTKB

5 Discussion

In this chapter, the previously reported results will be interpreted and placed into context with both each other and preexisting research, if applicable.

5.1 Generation Method

The discussed subjects of this section include the usability of the chosen scores for their respective tasks, a conclusion as to the model choice for this thesis, and the decision regarding the knowledge source for the prompts.

5.1.1 Score Usability

While CLIPScore and UQI have proven useful for other work (Abu Ahmad et al. (2023), Hessel et al. (2021), Krishnan et al. (2023), Chow et al. (2016)), their usability for comparing the suitability of approaches in this work had to be investigated. For this, additional UQI and CLIPScores have been documented for subsets of the Regular dataset. These scores, as well as the Regular and Random scores, can be seen in Fig. 5.1.

Dataset	KS	CLIP Mean $\pm$ SD	UQI Mean $\pm$ SD
Regular	KBp	91.4 ± 2.5	0.46 ± 0.24
	LLMp	91.5 ± 2.8	0.47 ± 0.24
	VLMp	91.5 ± 3.2	0.51 ± 0.25
Random	KBp	90.2 ± 2.8	0.49 ± 0.19
	LLMp	90.4 ± 2.8	0.47 ± 0.18
	VLMp	89.9 ± 3.1	0.55 ± 0.21
Shuffle	KBp	87.1 ± 2.5	0.43 ± 0.24
	LLMp	87.1 ± 2.4	0.43 ± 0.22
	VLMp	87.3 ± 2.3	0.51 ± 0.24
Category-Shuffle	KBp	88.9 ± 2.1	0.47 ± 0.26
	LLMp	88.6 ± 1.7	0.48 ± 0.23
	VLMp	88.7 ± 2.2	0.51 ± 0.25

Table 5.1: CLIP and UQI Scores by dataset and knowledge source (KS) for Dall-E 2

The Shuffle subset was created by randomly shuffling the subjects for the generated images of the Regular dataset. Any subject, except for the original, could be assigned to any image. Category Shuffle had a similar approach, but a subject could only be assigned to the image of a subject within the same category (as seen in Tab. 3.1 and 3.2). Through this, it could be tested whether and to what extent UQI and CLIPScores varied for images generated with this thesis's approach.

For UQI, all images had scores around 0.50 and standard deviations around 0.20. As UQI is a pixel-based quality index (Zhou Wang and Bovik (2002)) mostly tested with image-distortion pairs, this was within expectations. It cannot judge semantic accuracy or adequacy, but compares the literal pixel values of two images. This meant that even if two images showed the exact same entity, if, for example, the lighting in the images differed vastly, the UQI would score them similarly to different subjects with the same discrepancy in lighting. However, previous tests of image generation from knowledge base triples yielded some heavily distorted images like Fig. 5.1 (image from a test for the research project preceding this thesis). For that reason, the UQI was chosen to remain in this work to filter out such unusable images without using computationally more expensive metrics like CLIPScore. The observed value ranges for the UQI, staying around 0.50, are similar to the scores received for a random cat image compared to another natural image. This can be understood as the generated images' quality being closer to that of a natural image or photograph than that of a pure black image, indicating that the generated images possess a good aesthetic quality. Since the quality of a generated image as a natural image is not dependent on the subject, it is expected to see very similar UQI scores across all tests and subjects.

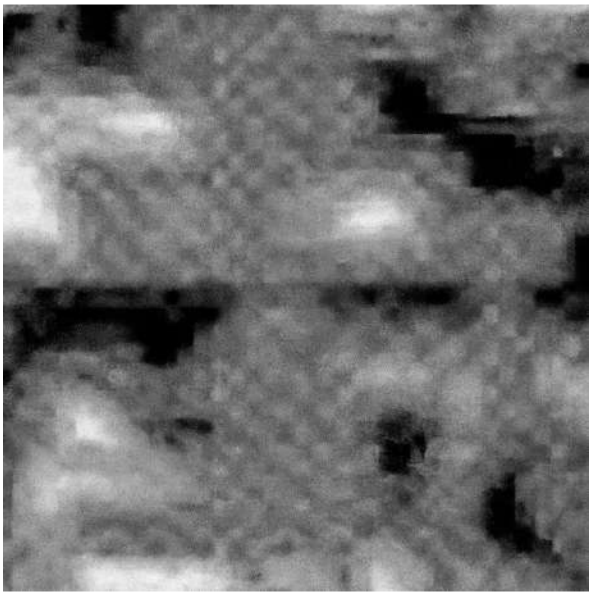


Figure 5.1: Heavily distorted image generated from triples for the book series "A Song of Ice and Fire" from a test for the research project preceding this thesis

For CLIPScore, the trends in the data are clear. The handpicked subjects of the Regular dataset performed the best (91.4–91.5). Randomly chosen subjects have comparatively lower CLIPScores (89.9–90.4), even though they still used the correct pairing of subject and generated image. It could be that the CLIPScore simply varies in value across different subjects, making it impossible to judge image adequacy through it. Because of this thought, the Shuffle and Category-Shuffle were evaluated. Both scores were even lower than the Random dataset scores (Shuffle: 87.1–87.3, Category-Shuffle: 88.6–88.9). These scores are very close to the comparison of an image with a completely black image for Shuffle and the

comparison of an image with an unrelated image for Category-Shuffle.
This means that the images were judged as more appropriate in comparison to a reference image when subjects were shuffled within their own category, than when they were shuffled within all available subjects. Since both shuffle sets were still assessed as lower in representative quality than the Random dataset, the CLIPScore can be assumed to accurately model image adequacy in comparison to a reference image. This means image adequacy decreased from handpicked to randomly chosen subjects, even though both had the generated image of a subject compared to a reference image of the same subject. Since all handpicked subjects could be considered famous or popular subjects, while a random choice would choose any subject regardless of popularity, this leads to the hypothesis that the popularity of a subject influences the quality of its generated image. This hypothesis, congruent with RQ 3, will be addressed in a later section.

5.1.2 Used Model

At this point in time, many different types of vision models exist and can generate very realistic-looking images (Midjourney (2022), Gu et al. (2022), Rombach et al. (2022), Meta (2025b)). However, even though the results are generally considered high-quality, there are still differences in quality between models (Li et al. (2024)), especially for specific tasks like generating photorealistic faces (Borji (2023)) or data visualization (Ye et al. (2024)). In addition, models vary in pricing and availability. For this work, Dall-E 2 (Ramesh et al. (2022)) and a Stable Diffusion model (stable-diffusion-3.5-large-turbo, Stability AI (2024a)) were available. To decide between them, the costs and generation quality were compared.

In terms of costs, the Stable Diffusion model could not be beaten for this work. Due to Stability AI's stable-diffusion-3.5-large-turbo model being publicly available, no costs other than for hosting were accumulated. Since the hosting was not bound to this work alone but to multiple other experiments, these costs can be neglected for this consideration.
As previously stated in 3.3.1, Dall-E 2 had generation costs of $ 0.02 per generated image[1]. Even without considering the estimate of $ 6,100 for generating images for 305,000 entries (around 5 % of GPTKB v1.5.1), any costs are higher than the ones necessary for the Stable Diffusion model. Therefore, the only reason to consider Dall-E 2 for this work would be a drastically better performance for generating representative images.

The actual quality of images in regard to factors like blurriness and contrast, as determined by the UQI scores, did not vary between Dall-E 2 and the Stable Diffusion model. Both were around 0.50 with a standard deviation of about 0.20. Concerning the image adequacy, how well an image fits the subject it is portraying, as judged through the CLIPScore, though, the models differed slightly. The Stable Diffusion model consistently acquired higher CLIPScores throughout all experiments, even performing similarly well for the Random dataset as Dall-E 2 did for the Regular dataset. This means the images generated via the Stable Diffusion model were semantically closer to the gathered reference images than the images generated via Dall-E 2.
A reason for that might be guardrails that commercial models like Dall-E 2 have to prevent misuse, taking various shapes like blocking individual words or phrases, or having prompts evaluated through an LLM (Villa et al. (2025)). Though it is technically possible to bypass these safeguards (Jiang et al. (2025)), this is against OpenAI's user guidelines. For prompts containing blocked words or phrases, causing a "Bad Prompt Error" for most cases or an "Internal Server Error" for all monkeys and apes, the only way to generate an image is to rephrase the prompt. Since replacing potentially problematic terms with paraphrases

[1] as of April—September 2025

makes the prompt less specific, continuous paraphrasing can alter the content of a prompt significantly. Words and phrases identified as necessitating paraphrasing in this work were all monkeys and apes, current, famous people, historic people who are generally considered problematic, as well as words with sexual connotations like "rubber". Due to the common appearance of these words and phrases in a knowledge base, paraphrases were needed on 28 occasions for the Regular dataset as generated by Dall-E 2, possibly reducing the image adequacy because of it.
The recognizability of images, as identified through the guessing game, did not seem to be as affected as the CLIPScores. Still, the Stable Diffusion model outperformed Dall-E 2 in most cases, but the differences were relatively small, especially considering the standard deviations between the 6 tests of each model.

Manual review of images yielded similar results, with a preference for the images generated by the Stable Diffusion model, in both recognizability and quality. This answers RQ 1, showing that the choice of model does impact the resulting images' recognizability and quality, though to only a small degree when comparing Dall-E 2 and the Stable Diffusion model.
In conclusion, the Stable Diffusion model performed mostly better than Dall-E 2 by a small margin and had much lower costs associated with its use. Therefore, it was chosen as the model to use for the larger-scale tests and the final, large-scale generation at the end of this work.

5.1.3 Prompts

After choosing a model, the knowledge source used for generating a prompt had to be decided. The scores for the three knowledge sources were very similar when compared across all generated images of a dataset. When looking at the scores for the chosen model, the Stable Diffusion model, an interesting observation can be made. For the 120-subject datasets, Regular and Random, the KB prompt has the worst performance, being 0.4 and 0.6 behind the best knowledge source in those cases, LLMp. However, in the larger randomized dataset, New Random, it has the best performance, at 0.8 above both other options.
Additionally, popularity affected the rankings of the knowledge sources. For the Stable Diffusion model, KBp had an average CLIPScore change from popular to long-tail subjects of -1.47 (-0.2 for Regular, -2.3 for Random, -1.9 for New Random). The other knowledge sources, in comparison, showed changes of -1.63 (-1.3 for Regular, -1.6 for Random, -2 for New Random) for LLMp and -1.57 (-1.9 for Regular, -2.2 for Random, -0.6 for New Random) for VLMp. From this, it can be inferred that KBp is the best choice of knowledge base for long-tail subjects, since it loses the least amount of image adequacy. Since long-tail subjects, by definition, make up most of the items within a knowledge base, their generation quality should weigh more for the choice of knowledge source. In addition to the higher weight due to long-tail subjects, the next best knowledge source in terms of a difference between popularities, VLMp, showed lower CLIPScores for the New Random dataset, which is more similar to the larger-scale goal than the Regular and Random datasets. Its scores for the New Random dataset were lower for both popular and long-tail subjects, the difference for popular and long-tail subjects being as large as 1.9, and its score for popular subjects was the same value as the score for the long-tail subjects using the KB prompt.

Weighing these facts against each other, it was decided to proceed with the KB prompts for further tests and the large-scale generation at the end of this thesis, answering RQ 2. Using natural language prompts generated from the triples of knowledge base entries, like this, has gotten great results for Abu Ahmad et al. (2023) as well, further supporting the choice.

5.2 Scalability

This section will discuss and interpret the results regarding the scalability of the image generation approach used in this work to illustrate knowledge base entries. The parts looked at in detail are the impact of subject popularity, the feasibility of reuse, and the question of the generation speed.

5.2.1 Subject Popularity

As the results of this thesis have shown, the popularity of the portrayed subject has an impact on the image adequacy, as shown through the CLIPScore, answering RQ 3. Since popular entities are more likely to have more prevalence in training data due to larger amounts of data available for them overall, the vision models being able to more accurately illustrate them makes sense. Additionally, for the same reason, popular subjects tend to have longer and more correct entries in GPTKB, leading to more detailed and clear prompts. The entries and correlating prompts of the subjects "1 Bethel Valley Road, Oak Ridge, Tennessee" and "Toucan" are an example of this phenomenon.
The entry in GPTKB for a random address located in the US, "1 Bethel Valley Road, Oak Ridge, Tennessee", has a total of 7 triples associated with it, in contrast to the well-known bird "Toucan" with 30 triples in its entry. As seen in Fig. 5.2, this causes the prompts to show large differences in their specificity. The prompt for "1 Bethel Valley Road, Oak Ridge, Tennessee" describes a generic, American house in a just as generic scenery, leaving the vision model to fill in the blanks from its own knowledge. In contrast, the prompt for the more popular subject, "Toucan", includes visual details like a branch in a lush, tropical forest, or the large, colorful bill, since it could gather this information from the longer knowledge base entry.
This discrepancy then gets reflected in the generated images, causing lower scores for long-tail subjects. The more unknown the subject is, the less information is in the knowledge base entry, and the more the vision model has to improvise upon very generic prompts. Since the vision model most likely has less information about long-tail subjects as well, though, it mostly stays true to the prompt, creating an image fitting the very broad description.

```
Generate a realistic image of a building or house located at 1 Bethel Valley Road, Oak Ridge, Tennessee, set against a backdrop of natural scenery, incorporating elements that reflect the region's architecture and surroundings, in a style that emphasizes clarity and geographical accuracy.
```

(a) The prompt for "1 Bethel Valley Road, Oak Ridge, Tennessee"

```
Generate an image of a brightly colored Toucan perched on a branch in a lush tropical forest, showcasing its distinctive large and colorful bill, in a vibrant and realistic style, highlighting its typical characteristics as a member of the Ramphastidae family and Piciformes order.
```

(b) The prompt for "Toucan"

Figure 5.2: Comparison of the prompts of a long-tail (the US-based address 1 Bethel Valley Road, Oak Ridge, Tennessee) and a popular (the well-known bird toucan) subject as used in the Large Scale dataset

It was shown earlier that the Random dataset scored worse than the Regular dataset for both models and both popularities. This could be explained by popularity as well. The handpicked subjects of the Regular dataset are all more or less well-known. Though subjects sorted into "long-tail", like "Kansas City" or "Sacramento", are substantially less well-known than ones sorted into "popular", like "Albert Einstein" or "Miami", they are still much more popular than even some popular subjects from the Random dataset, like "Ge Force GPU" or "Karpov vs Korchnoi, Game 19". Because of that, it is not surprising that the Regular dataset, with its overall more popular subjects, outperforms both the Random and the larger New Random datasets.

One part of the results seems to go against these interpretations, however: The scores for the long-tail subjects of the New Random dataset are better than those of its popular subjects. Multiple reasons could be the cause.
For one, the split into "long-tail" and "popular" was done via ChatGPT, saving time and labor, but inviting errors. Since it was only checked if all subjects were present and unique, the actual sorting might have had errors. It depends on what can truly be defined as "long-tail" and "popular", as well as how ChatGPT defined it during the sorting process. A subject considered "popular" by ChatGPT might have so little information in GPTKB or the training data of the Stable Diffusion model that it would be considered "long-tail" from their perspective.
Another reason could be the imbalance of popular and long-tail subjects. The way ChatGPT sorted the subjects, many more subjects were categorized as long-tail than popular. In fact, only 15 of the 200 randomly chosen subjects of the New Random dataset were considered popular by ChatGPT. This skewed distribution could have caused outliers to more heavily affect the scores of the very small subset for the popular subjects.
The last reason is a combination of the two, which is most likely. If even just a couple of actually long-tail subjects, with rather unrecognizable images, are mistakenly assessed as popular subjects, the heavily skewed distribution would cause their lower scores to heavily decrease the average score of all popular subjects. Though the Random dataset had just 15 subjects categorized as popular as well, its lower count of 120 subjects in general causes the skew to be lighter. Additionally, it might have simply included more popular "popular" subjects or fewer falsely sorted subjects, causing it to still show the expected behaviour of lower scores for long-tail versus popular subjects.

5.2.2 Reuse

Reusing images to reduce the number of images that have to be generated seems like an obvious approach when attempting to create illustrations for large numbers of subjects, like an entire knowledge base’s items. This thesis reports that the answer to RQ 4 is not straightforward. Comparing the changes in CLIPScore for the different Reuse cases, Reuse Company/City and Reuse Product/Landmark, makes it clear that reuse comes at the cost of some image adequacy. However, this cost is different, depending on the specific reuse cases. In the tests of this thesis, Reusing landmarks for their cities had worse results than reusing the cities for their landmarks. Conceptually, this does not make sense, since representing a whole by a part of it should work a lot better than representing a part of something by its whole. After all, using the specific case considered in this thesis, someone could guess "New York" from seeing the Statue of Liberty, but guessing "Statue of Liberty" from an image of New York seems counterintuitive. Considering that the city images were more adequate for their subjects than the landmarks for theirs, perhaps the images were still better even when reused. Another possibility would be that the city images already include the landmark, which would understandably make them an adequate image for the landmark too.

The calculated CLIPScores for products when using the images of the companies that make them had almost the same values as the opposite case. However, considering the decrease from the regular scores to the scores with reuse, reusing the product images for the companies (average decrease of 0.8) shows better results than reusing company images for their products (average decrease of 3.0). This might seem contradictory, but since the regular scores for the companies had relatively low CLIPScores compared to the other categories already, the company images themselves might be the key to understanding this phenomenon.
On the one hand, generating a representative image for a company from just text prompts is difficult. It needs either the name of the company to be displayed or to show a clear logo. The former was almost impossible for the images whose CLIPScores were compared, as those were generated with Dall-E 2, which seemed to struggle a lot with generating words in the images, as shown in Fig. 5.3. The latter would've relied on detailed, text data for the company logos, which the given GPTKB entries did not include. The only hope for the vision model was its own inherent knowledge of the company by its name alone, which can be difficult, especially for smaller or more local companies, most likely explaining the lower scores for the specific category, "Companies".

(a) Image generated for "Luxottica Group"   (b) Image generated for "Furukawa Co., Ltd."

Figure 5.3: Examples of words with unrecognizably scrambled letters generated on images by Dall-E 2

To look at the influence of reuse on the recognizability of images, the guessing game was used on the 500-subject subset of the Scale Test. While this did not result in a detailed exploration of different categories or individual score differences, it could give a good overview of how reuse could impact average image adequacy across a larger-scale generation. Both the false predictions and the standard deviation between the six runs of the guessing game rose, indicating a lower prediction confidence of Llama. This means that the image adequacy is affected, but since the changes were only minor (average of 14.8 % wrong predictions, up from 13.7 %), this work does not consider the influence too major to apply reuse.

The first experiment, comparing the separate categories, made it clear that, to get an even better grasp on how to optimally reuse images, the choice of which subject could be reused for another was of great importance. To retain scalability, the decision-making process regarding this had to be automated. Though it was considered to use a GPTKB inherent factor, like the bfsParent predicate, the idea was rejected. The predicates within GPTKB are too unpredictable by themselves, and some sort of ranking would have to be created to decide between multiple parent items. Instead, it was decided to compare the whole entries directly. Since creating text embeddings once and comparing those is a lot faster than comparing each possible pair of texts semantically (Reimers and Gurevych (2019)), text embeddings were used. Looking at how increasing reuse reduced the CLIPScore could grant an approximate threshold of how many images could be reused without decreasing the image adequacy too far. Considering the first sudden drop in the resulting curve between 20 and 30 %, setting the cutoff to about 25 % seems reasonable.

Lastly, for this topic, it should be mentioned that reuse has follow-up issues for large-scale generations with no set end time. As more and more subjects and therefore images get added, a new subject might have a similar entity with a similarity score higher than the threshold that is already using a reused image. Even though the entries are then similar enough for reuse, the actual entry to be considered would be the original one that created this image. It also means all entries, their text-embeddings, and the generated images for them have to be available for the running code, as well as a remark whether an image was a reused image and from which entry it hailed.

5.2.3 Generation Speed

A major part of scalability is the required time for generation. In the large-scale generation that was done at the end of this thesis, no reuse of images happened to receive a baseline of pure generation speed and create a base of generated images to build upon in the future. Before the generation process could start, the newest version of GPTKB was transformed from a CSV file to a SQLite database with a unique index over the subject to speed up the entry retrieval. This meant that each entry retrieval took under 0.1 s, which is negligible in comparison to the prompt and image generation process. The prompt generation varied between 0.1 and 3 s while the image generation kept around 2–3 s. Based on this, assuming an average of around 4.15 s per subject, the generation process for the final 104,611 images took around 120.6 h. In reality, the process took slightly longer, since generation times would go up, most likely whenever the APIs were used by other projects at the same time, sometimes even rising to over 70 s per generation.

If the optimal speed was used for all 6,100,000 entries of GPTKB 1.5.1, a bit over 290 days of constant generation would be required to create an image for all entities. Reusing the proposed amount of 25 % of images would mean only generating around 4,575,000 images, reducing the generation time to around 219 days. However, this does not include the additional time needed for finding the best reuse match and applying it. This time will also increase with each new subject added and illustrated, as each new entry has to be embedded first, the number of embeddings to compare increases, and the potential depth of the original subject for a reused image increases. To further reduce the generation time, the actual generation processes for prompts and images would have to be minimized.

5.3 Implications and Challenges

This section will interpret the results regarding the value provided by the generated images, as well as challenges during the thesis, like potential biases of the used models influencing the created data and logical errors within the generated images.

5.3.1 Image Worth

To answer RQ 5 and find the worth of a generated image for a knowledge base entry in comparison to triples, the predictions in the guessing game using images were compared to the predictions in the guessing game using tuples from the knowledge base entry. That way, an objective measurement could be done to directly compare the value for the recognizability of an image and the first n predicate-object tuples. As seen in the direct comparison in Fig. 5.4, the prediction accuracy for the first tuple within the GPTKB entry approximates half of the performance of a generated image. This can be interpreted as an image's value being approximately half of a tuple, but certain considerations must be taken into account.

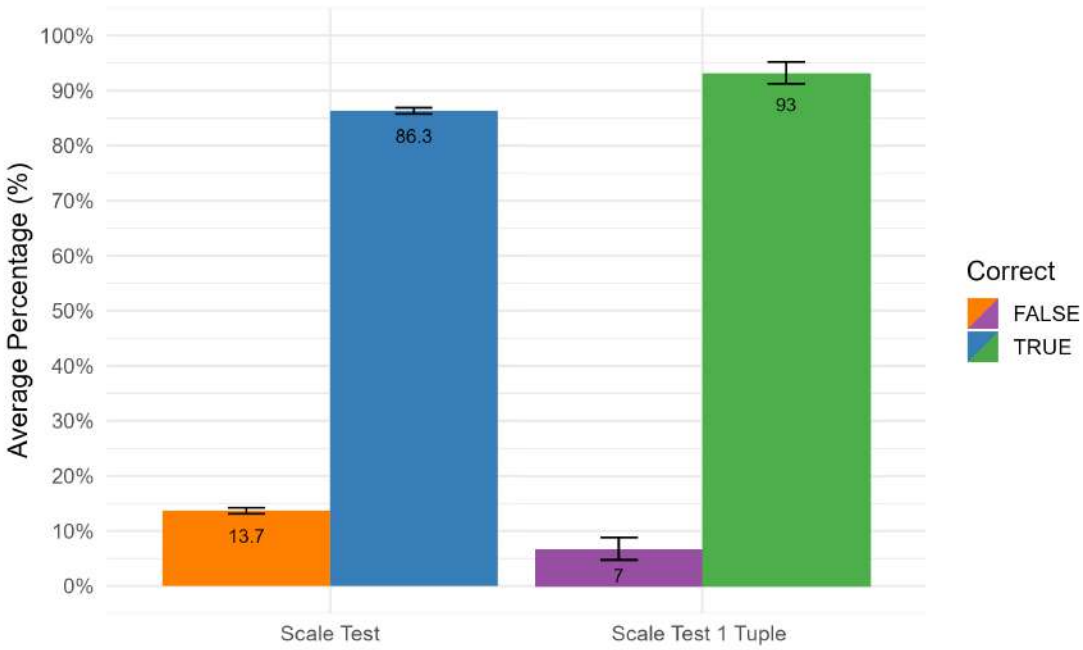


Figure 5.4: Guessing game scores for 500-subject Scale Test subset for images and 1 tuple

Firstly, this comparison is based on the recognizability of the subject for an AI. A human's perception might differ, though it is difficult to estimate whether they would have an easier or harder time identifying the subject from the image versus the tuple. Identifying a subject from a single tuple can be considered equivalent to a zero-shot entity linking task, as introduced by Logeswaran et al. (2019), though the use case of this thesis had the added guess options, simplifying the task slightly. Most tuples extracted from GPTKB can give a good chance to narrow down the options to just the ones that fit that specific description. Essentially, as long as the guess options are different enough from each other, even a single, randomly chosen tuple will be enough to identify the correct subject. If multiple subjects of the same or a similar "instanceOf", for example, a "village" and a "city" or two subjects with "instanceOf person", the correct subject becomes more ambiguous. However, because the images of two similar entities like that would be similar as well, as shown in Fig. 5.5, it is still difficult to tell whether a human would have more trouble with the images or the tuples. This coins the unproven hypothesis that, for a human, the value of the generated image would be similar to the first tuple as well. At the same time, it is known that humans and AI show clear differences in how they recognize subjects from images (van Dyck et al. (2021)), especially if the subjects are portrayed in unusual poses (Ollikka et al. (2025)). This could potentially increase the value of an image for a human, but it is impossible to accurately predict without further study.

(a) "North Grafton, Massachusetts", "instanceOf", "village"

(b) "Bend, Oregon", "instanceOf", "city"

Figure 5.5: Images from Scale Test and first triples (subject, predicate, object) for "North Grafton, Massachusetts" and "Bend, Oregon" as examples for similar subjects that appeared together as guess options

Secondly, while it is a comparison of a generated image's worth with that of a randomly chosen tuple, the value of individual predicate-object tuples most likely varies. This means, an image's value might be around half that of a tuple on average; in comparison to certain types of tuples, the worth might be higher or lower. To investigate this, more studies using a format like the guessing game with tuples would have to be performed. Preferably, it would use different types of tuples from various entries, as well as checking the value for popular and long-tail subjects individually.

Thirdly, since the tests to determine the image worth were performed with the Scale Test dataset, the dataset itself could have influenced the results. Specifically, the Scale Test dataset was generated from the first 5,000 subjects of GPTKB sorted by their bfsLayer. Earlier generated subjects have a lower bfsLayer, with the first item, Vannevar Bush, having the bfsLayer 0, and later subjects have a higher bfsLayer. More specifically, all items generated as child-subjects of Vannevar Bush have the bfsLayer 1, all generated from those the bfsLayer of 2, and so on. This makes it more likely to find more popular items within the lower bfsLayers, as those have more immediate links to the well-known starting subject, Vannevar Bush. It creates the possibility that true long-tail subjects were not represented, especially in the only 500-subject large, randomly chosen subset that was used for the test, to reduce runtimes. It could be that less well-known, more long-tail subjects would cause worse prediction accuracies for the images, tuples, or both.

5.3.2 Biases in GenAI

Regarding RQ 6, about notable biases in the generated data, the results were quite clear for depicted people. The ethnicity "Caucasian" was a lot more prevalent in the generated images than it was inferred to exist in the reference data from GPTKB, with an increase of 18.4 % if "Hispanic/Other" is not considered Caucasian and an increase of 10.8 % if it is. The increased occurrences of white people in the generated images are nothing new. In fact, it has been shown many times before that AI, especially vision models, output data with strong biases with regard to traits like ethnicity and gender (Park (2024), Bianchi et al. (2023), Zhou et al. (2024)).

Another obvious difference between the observed and referenced data is the stark difference in numbers for "Black" and "African". In the observed, "Black" only holds a prevalence of 0.4 % and "African" 9.8 %, in the referenced, "Black" has 7.2 % and "African" isn't present at all. This, however, can be attributed to the fact that something like ethnicity is almost impossible to accurately identify from an image, as Llama-4 was asked to do. "Black" and "African" could easily be used interchangeably as descriptions for a person in an image with no other identifiable features. Therefore, these numbers should be compared as sums, leaving the observed data with a slightly increased frequency of occurrence for "Black + African" at 10.2 % compared to the referenced 7.2 %.
The split between the different Asian ethnicities (categorized by Llama as "Asian", "East Asian", "South Asian", and "Indian") can be viewed similarly. For the same reason, it makes sense to compare the total percentages for them as well. This leaves the observed data at an "All Asian" percentage of 9.4 % in comparison to the referenced amount of 16.9 %, much outweighing the percentage within the generated images. Once again, this fits into the previously described biases of VLMs, pushing Western ethnicities and decreasing the prevalence of others, especially minorities.
It should be kept in mind that the reference values were only inferred through the 11 most common nationalities in GPTKB, and the official counts and numbers of those countries as gathered from Wikipedia (2025). They do not perfectly reflect the true distributions of ethnicities in GPTKB or the real world.

The age analysis had no reference data, as ages are not commonly reported for the people within GPTKB. It does show two larger camps of age groups. One at 21–30 and three similarly sized ones at 41–50, 51–60, and 61–70. Together they all make up over 3/4 of all ages generated, though the three older, connected age groups make up over 3.6 times as many of the images as the 21–30 group. In combination with the strong bias toward men, generating them approximately 1.6 times as often as they appear in the reference material of GPTKB, this might correspond with the findings of Zhou et al. (2024). They found women to be more often generated in younger ages, and men in older ages. If more men were generated, and those tend to be generated at older ages, this would explain the skewed distribution of ages. At the same time, the differences between the three age groups between 31 and 70 are minor. It could be because the visual distinction between people within the age groups from each other is fluid, but it is difficult to determine the true reason, as that would necessitate understanding the decision-making process of Llama during the age prediction.

Figure 5.6: Images for "American Society of Magazine Editors" with a group of people used for the bias check as generated by the Stable Diffusion model for the Large Scale dataset

All in all, though, there are some points to keep in mind for this bias analysis. For one, it does not accurately portray the biases of vision models in general, since the comparison was done with GPTKB, an AI-generated knowledge base that most likely has some biases in and of itself. For another, the bias check was done with a 500-image subset from around 100,000 generated images. This means percentages of 500 images were compared with the percentages of over 800,000 entries in GPTKB. The relatively low number of observed images could mean the percentages are more influenced by the random choices of the 500 images, possibly showing a different distribution at higher counts of images analysed. Additionally, the images were chosen after Llama-4 checked if the image included "a central person in the image who has a visible face", choosing yes if someone could guess traits like age or ethnicity from the image, and no otherwise. Again, in the actual trait-guessing process, only the central person's traits were supposed to be guessed. This caused the inclusion of images like Fig. 5.6, in which only one of the people had their traits guessed by Llama-4.

Interestingly, the central person Llama-4 chose seemed not always to be the literally most central person, as can be seen from the guess for "California Suite" as seen in Fig. 5.7a. For this, it guessed "Male" even though the physically most central person of the image reads very clearly as female. This could just be a mistake in Llama's vision, but it seems unlikely, since it happened repeatedly, as shown through Fig. 5.7b. Due to the repeated occurrences, a bias seems more likely as a cause than coincidentally misidentifying the gender of the central person as "Male" in multiple mixed-group images.

(a) Image of "California Suite"; Guessed traits: Caucasian, 50, Male

(b) Image of "Flirting with Disaster"; Guessed traits: Caucasian, 30, Male

Figure 5.7: Images of groups with a central woman guessed as male for the bias check as generated by the Stable Diffusion model for the Large Scale dataset

5.3.3 Logical Errors

It is well known that, while GenAI is able to produce highly realistic images, logical errors like fused animals or objects, too many fingers, or trains without railways are common in the generated images as they are the result of AI hallucinations (Aithal et al. (2024)). To check the final results of this thesis for such errors, a manual review was done. 500 images were randomly chosen from the Large Scale dataset, reviewed manually, and either counted as being free of logical errors or not.

It should be specified that "free of logical errors" in this context did not correspond with "There were no artifacts or illogical parts in the image whatsoever.". Instead, it was considered "close enough to no errors" that a viewer would be left in suspension of disbelief, and consider the image logical as a whole even with these errors (Ganczarek et al. (2015), McGowan (2024)).

Keeping this in mind, 182 of the 500 images (36.4 %) were considered "error-free", leaving 318 (63.6 %) with errors major enough to break any illusion of an actual photograph or human-made illustration. The major errors were almost exclusively anatomical errors, like too many fingers, melted fingers and hands, additional legs, only partially rendered faces, and fused animals or objects. Some other reasons for being judged as too error-heavy were jumbled words making up an important part of the picture, like clearly visible names of buildings or book titles. Examples of an error can be seen in Fig. 5.8a. None of the randomly chosen images that had railways of any sort passed the test, as they left the vehicles completely next to the rails or rails merged into each other or the ground, or all of the former. Examples for this can be seen in Fig. 5.8b.

(a) Image of "Anas rubripes", with seemingly two ducks merged into one

(b) Image of "Holyoke station", with train next to the track and rails disappearing into nothingness

Figure 5.8: Images generated by the Stable Diffusion model for the Large Scale dataset counted as having logical errors

The images counted as "error-free" mostly consisted of human portraits that do not show hands (Fig. 5.9a), as well as wide shots of land- and cityscapes (Fig. 5.9b), rivers with only small humans, and all shots that had more detailed things like human masses that were so small, they were too blurry to expect details.

(a) Image of "Harriet Barrow"

(b) Image of "Yankton County, South Dakota"

Figure 5.9: Images generated by the Stable Diffusion model for the Large Scale dataset counted as free of logical errors

Bibliography

Raia Abu Ahmad, Martin Critelli, Şefika Efeoğlu et al. 2023. Draw Me Like My Triples: Leveraging Generative AI for Wikidata Image Completion. In *CEUR Workshop Proceedings*, Athenes, Greece.

Mohanad Abukmeil, Stefano Ferrari, Angelo Genovese et al. 2021. A Survey of Unsupervised Generative Models for Exploratory Data Analysis and Representation Learning. *ACM Comput. Surv.*, 54(5).

Sumukh K. Aithal, Pratyush Maini, Zachary C. Lipton and J. Z. Kolter. 2024. Understanding Hallucinations in Diffusion Models through Mode Interpolation. *Advances in Neural Information Processing Systems*, 37:134614–134644.

Jens Peter Andersen, Lise Degn, Rachel Fishberg et al. 2025. Generative Artificial Intelligence (GenAI) in the research process – A survey of researchers' practices and perceptions. *Technology in Society*, 81:102813.

Jason Ansel, Edward Yang, Horace He et al. 2024. PyTorch 2: Faster Machine Learning Through Dynamic Python Bytecode Transformation and Graph Compilation. Publication Title: 29th ACM International Conference on Architectural Support for Programming Languages and Operating Systems, Volume 2 (ASPLOS '24) original-date: 2016-08-13T05:26:41Z.

Anthropic. 2023. https://www.claude.com/product/overview.

Sören Auer, Christian Bizer, Georgi Kobilarov et al. 2007. DBpedia: A Nucleus for a Web of Open Data. In *The Semantic Web*, pages 722–735, Berlin, Heidelberg. Springer.

Janet Beagle. 2009. Pictures versus text: Modality effects across three levels of learning and study time. *ETD Collection for Purdue University*.

James Betker, Gabriel Goh, Li Jing et al. 2023. Improving image generation with better captions. *Computer Science. https://cdn. openai. com/papers/dall-e-3. pdf*, 2(3):8.

Federico Bianchi, Pratyusha Kalluri, Esin Durmus et al. 2023. Easily Accessible Text-to-Image Generation Amplifies Demographic Stereotypes at Large Scale. In *2023 ACM Conference on Fairness, Accountability, and Transparency*, pages 1493–1504. Comment: FAccT 2023 paper. The published version is available at 10.1145/3593013.3594095.

Ali Borji. 2023. Generated Faces in the Wild: Quantitative Comparison of Stable Diffusion, Midjourney and DALL-E 2. *Preprint*, arXiv:2210.00586. Comment: dataset link udated!

Zhiyuan Chen. 2018. https://icrawler.readthedocs.io/en/latest/index.html.

Zhiyuan Chen. 2025. https://github.com/hellock/icrawler. Original-date: 2016-03-19T12:12:36Z.

Terry L. Childers and Michael J. Houston. 1984. Conditions for a Picture-Superiority Effect on Consumer Memory. *Journal of Consumer Research*, 11(2):643–654.

Li Sze Chow, Heshalini Rajagopal and Raveendran Paramesran. 2016. Correlation between subjective and objective assessment of magnetic resonance (MR) images. *Magnetic Resonance Imaging*, 34(6):820–831.

Liangping Ding, Cornelia Lawson and Philip Shapira. 2025. Rise of Generative Artificial Intelligence in Science. *Scientometrics*, 130(9):5093–5114.

Wenqi Fan, Yujuan Ding, Liangbo Ning et al. 2024. A Survey on RAG Meeting LLMs: Towards Retrieval-Augmented Large Language Models. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, KDD '24, pages 6491–6501, New York, NY, USA. Association for Computing Machinery.

Shuai Fu, Jian Zhou, Qi Chen et al. 2025. Counting Hallucinations in Diffusion Models. *Preprint*, arXiv:2510.13080.

Isabel O. Gallegos, Ryan A. Rossi, Joe Barrow et al. 2024. Bias and Fairness in Large Language Models: A Survey. *Computational Linguistics*, 50(3):1097–1179.

Joanna Ganczarek, V. Ruggieri, D. Nardi and M. Olivetti Belardinelli. 2015. Intersection of reality and fiction in art perception: Pictorial space, body sway and mental imagery. *Cognitive Processing*, 16(1):233–236.

GPTKB. 2025. https://gptkb.org/entity/Vatican_Apostolic_Palace/.

Jiaxi Gu, Xiaojun Meng, Guansong Lu et al. 2022. Wukong: A 100 Million Large-scale Chinese Cross-modal Pre-training Benchmark. In *Advances in Neural Information Processing Systems*, volume 35, pages 26418–26431. Curran Associates, Inc.

Jie Gui, Zhenan Sun, Yonggang Wen et al. 2020. A Review on Generative Adversarial Networks: Algorithms, Theory, and Applications. *CoRR*, abs/2001.06937.

Stefan Hajkowicz, Conrad Sanderson, Sarvnaz Karimi et al. 2023. Artificial intelligence adoption in the physical sciences, natural sciences, life sciences, social sciences and the arts and humanities: A bibliometric analysis of research publications from 1960-2021. *Technology in Society*, 74:102260.

Jack Hessel, Ari Holtzman, Maxwell Forbes et al. 2021. CLIPScore: A Reference-free Evaluation Metric for Image Captioning. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 7514–7528, Online and Punta Cana, Dominican Republic. Association for Computational Linguistics.

William E. Hockley. 2008. The picture superiority effect in associative recognition. *Memory & Cognition*, 36(7):1351–1359.

Yujia Hu, Tuan-Phong Nguyen, Shrestha Ghosh et al. 2025a. GPTKB v1.5: A Massive Knowledge Base for Exploring Factual LLM Knowledge. *arXiv preprint*. Other 7 pages, 6 figures, 1 table
Version Number: 1.

Yujia Hu, Tuan-Phong Nguyen, Shrestha Ghosh and Simon Razniewski. 2025b. Enabling LLM Knowledge Analysis via Extensive Materialization. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 16189–16202, Vienna, Austria. Association for Computational Linguistics.

Zhengyuan Jiang, Yuepeng Hu, Yuchen Yang et al. 2025. Jailbreaking Safeguarded Text-to-Image Models via Large Language Models. *Preprint*, arXiv:2503.01839.

Andrew Khalel. 2025. https://github.com/andrewekhalel/sewar. Original-date: 2018-08-23T09:11:28Z.

Palani Thanaraj Krishnan, Parvathavarthini Balasubramanian, Vijay Jeyakumar et al. 2023. Intensity matching through saliency maps for thermal and visible image registration for face detection applications. *The Visual Computer*, 39(10):4529–4542.

Sean Lee, Aamir Shakir, Darius Koenig and Julius Lipp. 2024. https://www.mixedbread.ai/blog/mxbai-embed-large-v1.

Baiqi Li, Zhiqiu Lin, Deepak Pathak et al. 2024. Evaluating and Improving Compositional Text-to-Visual Generation. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 5290–5301.

Xianming Li and Jing Li. 2024. AoE: Angle-optimized Embeddings for Semantic Textual Similarity. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1825–1839, Bangkok, Thailand. Association for Computational Linguistics.

Lajanugen Logeswaran, Ming-Wei Chang, Kenton Lee et al. 2019. Zero-Shot Entity Linking by Reading Entity Descriptions. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 3449–3460, Florence, Italy. Association for Computational Linguistics.

Lu Ma, Kaibo Cao, Hao Liang et al. 2025. Evaluating and Predicting Distorted Human Body Parts for Generated Images. *Preprint*, arXiv:2503.00811. Comment: 8 pages, 6 figures.

Clara McGowan. 2024. *The Suspension of Disbelief Is a Prerequisite of the Immersive Experience in Contemporary Art*. Thesis, Institute of Art, Design + Technology.

Quintin McGrath. 2024. Responding to the Sharp Rise in AI in the 2023 SIM IT Trends Survey. *MIS Quarterly Executive*, 23(1).

Meta. 2025a. https://ai.meta.com/blog/llama-4-multimodal-intelligence/. Publication Title: Meta AI.

Meta. 2025b. https://huggingface.co/meta-llama/Llama-4-Scout-17B-16E-Instruct.

Midjourney. 2022. https://www.midjourney.com/home.

Dragoș M. Obreja, Răzvan Rughiniș and Daniel Rosner. 2025. Mapping the multidimensional trend of generative AI: A bibliometric analysis and qualitative thematic review. *Computers in Human Behavior Reports*, 17:100576.

Netta Ollikka, Amro Abbas, Andrea Perin et al. 2025. A comparison between humans and AI at recognizing objects in unusual poses. *Transactions on Machine Learning Research*, 2025(January):1–32. Publisher Copyright: © 2025, Transactions on Machine Learning Research. All rights reserved.

Open AI. 2025. https://openai.com/de-DE/index/image-generation-api/.

OpenAI. 2022. https://chat.openai.com. [Large language model].

OpenAI. 2025. https://openai.com/de-DE/index/gpt-5-system-card/.

OpenAI, Josh Achiam, Steven Adler et al. 2024. GPT-4 Technical Report. *arXiv preprint*. ArXiv:2303.08774 [cs]
Comment: 100 pages; updated authors list; fixed author names and added citation.

Ye Sul Park. 2024. White Default: Examining Racialized Biases Behind AI-Generated Images. *Art Education*, 77(4):36–45.

Heiko Paulheim. 2016. Knowledge graph refinement: A survey of approaches and evaluation methods. *Semantic Web*, 8(3):489–508.

Aditya Ramesh, Prafulla Dhariwal, Alex Nichol et al. 2022. Hierarchical Text-Conditional Image Generation with CLIP Latents. *arXiv preprint*. ArXiv:2204.06125 [cs].

Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics.

Robin Rombach, Andreas Blattmann, Dominik Lorenz et al. 2022. High-Resolution Image Synthesis with Latent Diffusion Models. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.

Shahnewaz Karim Sakib and Anindya Bijoy Das. 2024. Challenging Fairness: A Comprehensive Exploration of Bias in LLM-Based Recommendations. In *2024 IEEE International Conference on Big Data (BigData)*, pages 1585–1592.

Stability AI. 2024a. https://huggingface.co/stabilityai/stable-diffusion-3.5-large-turbo.

Stability AI. 2024b. https://stability.ai/news/introducing-stable-diffusion-3-5.

Georg Stenberg. 2006. Conceptual and perceptual factors in the picture superiority effect. *European Journal of Cognitive Psychology*, 18(6):813–847.

Fabian M. Suchanek, Gjergji Kasneci and Gerhard Weikum. 2007. Yago: A core of semantic knowledge. In *Proceedings of the 16th International Conference on World Wide Web*, pages 697–706, Banff Alberta Canada. ACM.

Gemini Team, Rohan Anil, Sebastian Borgeaud et al. 2025. Gemini: A Family of Highly Capable Multimodal Models. *Preprint*, arXiv:2312.11805.

Hugo Touvron, Thibaut Lavril, Gautier Izacard et al. 2023. LLaMA: Open and Efficient Foundation Language Models. *arXiv preprint*. ArXiv:2302.13971 [cs].

Leonard Elia van Dyck, Roland Kwitt, Sebastian Jochen Denzler and Walter Roland Gruber. 2021. Comparing Object Recognition in Humans and Deep Convolutional Neural Networks—An Eye Tracking Study. *Frontiers in Neuroscience*, 15:750639.

Vatican Tickets. 2025. https://vatican.rome-tickets.org/de/apostolischer-palast/.

Corban Villa, Shujaat Mirza and Christina Pöpper. 2025. Exposing the Guardrails: Reverse-Engineering and Jailbreaking Safety Filters in DALL·E Text-to-Image Pipelines. In *34th USENIX Security Symposium (USENIX Security 25)*, pages 897–916.

Denny Vrandečić and Markus Krötzsch. 2014. Wikidata: A free collaborative knowledgebase. *Communications of the ACM*, 57(10):78–85.

Xiahua Wei, Naveen Kumar and Han Zhang. 2025. Addressing bias in generative AI: Challenges and research opportunities in information management. *Information & Management*, 62(2):104103.

Gerhard Weikum, Xin Luna Dong, Simon Razniewski and Fabian Suchanek. 2021. Machine Knowledge: Creation and Curation of Comprehensive Knowledge Bases. *Foundations and Trends® in Databases*, 10(2-4):108–490.

Wikidata. 2025. https://www.wikidata.org/w/index.php?title=Q145093. [Online; accessed 24-October-2025].

Wikipedia. 2025. https://en.wikipedia.org/wiki/List_of_countries_by_ethnic_groups. Page Version ID: 1304670880
Publication Title: Wikipedia.

Thomas Wolf, Lysandre Debut, Victor Sanh et al. 2020. Transformers: State-of-the-Art Natural Language Processing. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, pages 38–45, Online. Association for Computational Linguistics.

Yajing Xu, Zhiqiang Liu, Jiaoyan Chen et al. 2025. Multi-modal Knowledge Graph Generation with Semantics-enriched Prompts. *Preprint*, arXiv:2504.13631. Comment: Accepted by IJCNN 2025.

Yilin Ye, Jianing Hao, Yihan Hou et al. 2024. Generative AI for visualization: State of the art and future directions. *Visual Informatics*, 8(2):43–66.

Mi Zhou, Vibhanshu Abhishek, Timothy Derdenger et al. 2024. Bias in Generative AI. *Preprint*, arXiv:2403.02726.

Zhou Wang and A.C. Bovik. 2002. A universal image quality index. *IEEE Signal Processing Letters*, 9(3):81–84.

6 Appendix

Category	Entries
Historical Figures / Military Leaders	Filippo Strozzi the Younger; Bernhard of Carinthia; Brigadier General Janis Sprogis; Radomir Putnik; Henry Fairfax, 2nd Lord Fairfax of Cameron; General Nguyen Van Hinh; Guillaume, Count of Flanders; George Burnham Jr.; Joseph Sewall III; Michele A. Lee; Mary Fitz Roy, 4th Duchess of Richmond; Boris Hybner
Wars / Battles / Military Events	Second Anglo-Boer War; Battle of Håkonarstadir; Capture of Fukuoka Castle; Battle of Yavin; 29th Logistics Support Battalion; U. N. forces; Karpov vs Korchnoi, Game 19; The Battle of the Shining Walls; Congo Free State Administration; UN Security Council Resolution 1704; German Freikorps; 1958 Baghdad Pact withdrawal
Arts / Music / Performance	Agnès Baltsa; Ms. Pat; Un Verano Sin Ti (Deluxe); Symphonic Live; Sérgio Reis - Ao Vivo (2015); LCD Soundsystem - This Is Happening; The Style Council Big Band; Flair for Fashion; Perlman Plays Tchaikovsky; The New England Transcendentalists; Giorgio Graziani; Detective Philippe Rousselot
Literature / Philosophy / Publishing	Prabhat Prakashan; Dr. Richard N. Longenecker; Algari Rose; The Binti Universe; Bloomsbury Publishing Grenada; Sri Bhashya; Murakami vs. Murakami (2016); Richard Day; George M. Marakas; The Merchant of Venice (Broadway); The Immortal Hulk Vol. 10; Lemaître's redshift; Marta Lobo Antunes
Cities / Regions / Places	Ceres, California; Tlaquepaque; Malden City Hall; Pingtung County Government; Riga's Historic Centre; Zalaegerszeg Adventure Park; Kintai Bridge Park; Bahawalpur Technical Institute; Churfirsten; Sheffield to Brora Line; Cott Canada; St. Albans City School District; Principality of Antioch; Erfurt Hauptbahnhof
Government / Law / Politics	Governor of Colorado Territory; Cherokee Nation v. United States Department of the Navy; 567 U. S. 709; Ajman Free Zone Authority; Liberty Global Partnerships; Massachusetts Route 9 G; Ascanian coat of arms; Ammonius Hermiae; Palazzo Ca' Corner della Regina; Seguin Transit; Baron of the Isle of Benbecula
Media / Film / Television	Dennis the Menace Strikes Again; And Then We Danced; NBC Sports Washington's The Game Plan; Best Music at the BAFTA Games Awards 2016; Screamfest Horror Film Festival 2008; Captain America #42; K9999; Gisele Yashar; Wizarding Wireless Network; Vikram Aur Betaal; Cagayake! GIRLS; Bell Productions; Paramount D. Smith
Museums / Parks / Culture Sites	The Museum of Contemporary Art in Bakersfield; Matsuyama City Museum of Poetry; Busan Sajik Stadium; Ban Chiang archaeological site; the Benin ceremonial drum; fresco of the Sacred Snake; Tokyo Metropolitan University Athletic Association; Molly Meldrum Foundation; Molesworth Airfield; Valkenburg railway station; Hiroshima University School of Dentistry; Great River Station; Yokohama Hakkeijima Sea Paradise
Sports / Games / Competitions	Rally of Czech Republic; All Blacks Sevens; Chico the Jet Hawk; Puzzle Bobble Infinity; Rurouni Kenshin Kengeki Kenran 26; Brit Asia TV Music Awards; KSL City Mall; Lizarazu; Vladimir Stimac; Dodge Dart Rallye; Eddie Goldman
Science / Tech / Industry	GeForce GPU; Stila; Paligo; FENSTERBAU FRONTALE; Piano Sonata No. 19 in C major, Op. 2; the presidential seal of Mexico; H. C. McCulloch; Kordes' 'Sunsprite' rose; Thomas Saltonstall

Table 6.1: Categories for the subjects of the Random dataset (categories chosen and sorted via ChatGPT)

Category	Entries
Historical Figures / Political / Military	Jean Kinkaid; Sir Syed Aftab Ahmad; Madho Singh; Neil Dunn; Bishop Emeritus Séamus Hegarty; Starost of Samogitia; Winifred Frances Knox; Pauline Roccucci; Charles W. Harkness; Pius Langa; Anna Amalia, Countess of Nassau-Siegen; Pamela Hill; Wong Ting-kwong; Grant Sawyer; Richard Jones, 1st Earl of Ranelagh; Frances Contreras; Jennifer Jordan; Deputy Inspector Genevieve Isom; Giovanni Migliara; Florence Laing Kennedy; Serhiy Honcharenko; Emperador de toda España; Admiral Amédée Courbet
Geography / Places / Infrastructure	Medway Valley Walk; Maracas Waterfall; Dean Prior; Bad Kreuzen; Bogorodsk; Durbar Square; Launceston, Cornwall, England; Keller, Indiana; June Park; Seattle, Washington, USA; Dinefwr Borough Council; Chevry; Solna Municipality, Sweden; Great Abaco Island; Diamond Hill MTR Station; SE Flavel Street; River Kensey; Naoshima; McLain, Mississippi, USA; Northern California, United States; Umeå–Östersund; Lotnisko Chopina w Warszawie; Spruceton Trail; Astley Castle; Calaboose African American History Museum; Iosan; Lotnisko Chopina warszawie; Augusta, Georgia; Jamestown, Virginia; Zaandam; Nordvestpassagen, Gjöa-ekspeditionen 1903–1907; Chinbrook Community Centre; Bursa Cable Car; Noor Bank Metro Station; NY 9A; Chekhov, Russia
Technology / Computing / Transport	HAL Laboratory; Tridium; llvm-readobj; Honeywell MAXPRO; Environmental testing – Part 1: General and guidance; Backdoor.Winstar.AH; Optiver; Protected Management Frames (PMF); MF trains; Nokia 6800; BMT Broadway Line; Mumbai Metro Line 27; Google Developer Experts; Audiohammer Studios; ASME B107.26-2003; State Road 75; New Hampshire Route 286A; French National Railways; Gauss lunar crater; Russian Missile Design Bureau; Oldsmobile 98 Regency Touring Brougham; Maybach SW 38
Arts / Culture / Literature	The Delinquent Season; KRS-One – KRS-One; Hungarian fairy tales; The Caves of Mars; The Unicorn in Captivity; The Citadel of Chaos; The Rock of Tanios; Crostata di ricotta e visciole; The Art of Readable Code; A Natural History of New York City; Fly Your Dreams; Symphony No. 1 in E major, Op. 5; All That You Dream; Anna Karenina (1935 film); The Dark Tower; The Prisoner of the Devil; Fortitude: The D-Day Deception Campaign; Thief in the Night; Fortitude The D-Day Deception Campaign; The Confessions of a Holiday Camp; Frank Butcher; Hoppy Uniatz
Education / Science / Language	College of Business and Professional Studies; Baker Heart and Diabetes Institute; Hezhe language; English College for Catholic priests; Lahore College for Women University; FSU Student Legal Services; Dynamics of Prejudice; Maryam Nassir Zadeh; Serena Nik-Zainal; Acoma Pueblo people; Amt II (Economic Affairs); Lipsha Horowitz; Empetrum nigrum; white-eyed gull; Orseolia; South American indigenous peoples
Music / Entertainment / Media	Steinway Artists; I Wonder Where My Baby Is Tonight; Let Them Eat Flax; Foster the People – Don't Stop (Color on the Walls); Seth Lakeman Band; Chris Cain (self-titled album); Zula Sound Bar; L'ultimo guerriero; Ascenseur pour l'échafaud (soundtrack); Yu-Gi-Oh! Rush Duel: Saikyou Battle Royale!!; Social Outcast; Tales of the Sinestro Corps; Jack Hanna's Animal Adventures; Jamie Kerstetter; Cory Rooney; Zentralquartett; Manta Pack Back Bling; Tea Cups; Third Eye Blind (album); Vielklang Studios; Northwest Public Radio
Religion / Myth / History / Misc	Innsmouth Hardware Store; Pate Lucid; Tsuru; General Who Defeats Bandits; Sheo; Adam (biblical figure); House of Ayala; Amal al-Jarari; Amir Mann; Lord Camber's Ladies; Donald Wiseman; James Wallace (pirate); Archbishop of Corfu; Hugh MacDonald, 1st of Sleat; Aunt Aggie; Cemetery of the Priests; Taiwan Creative Content Agency; Kim Gyeong-hwa; Sisters of St. Francis, Oldenburg; Aassoun; Dioecesis Tusculana
Politics / Law / Government	Strategic Arms Reduction Treaty signed; Public Law 100-119; New York State of Health; Embassy of Poland in Budapest; State Reception Center; Liberal Party of Russia; Duke of Biržai and Dubingiai; Dvach; New York Herald building; Propaganda and Censorship during Canada's Great War; 20th Space Range Squadron; Ashley Horace Thorndike
Sports / Travel / Local Culture	Stourbridge Line; Walker Ranch Loop Trail; South Lake Lean-to; Camberwell Market; 1937 Donington Grand Prix; Women's Artistic Gymnastics All-Around; Fight of the Night: Manel Kape vs. Felipe dos Santos; FC Cambuur; Popatlal; Illari; Glicério; Hjemkomst Center; Garden of the Gods Observation Trail; Cataño–San Juan ferry route; Welsh Athletics Hall of Fame; Lambay Whiskey; Aldo Pedro Poy; 2012 All-Star Game
People / Other	Thomas Rastrick (son); Dave Alvin; Ramey Jackson; Daniel Riley; Anu Noorma; Kitchie Benedicto; Julie Manet; Geraldine Picaud; Mary Belin du Pont; Vicki Kimm; Jason Brown; Jane Greenwood; Enrico Coveri; Lindsey Pfaff; Francisco del Villar

Table 6.2: Categories for the subjects of the New Random dataset (categories chosen and sorted via ChatGPT)